

CLASS X - SCIENCE

LIFE PROCESSES



LIFE PROCESSES

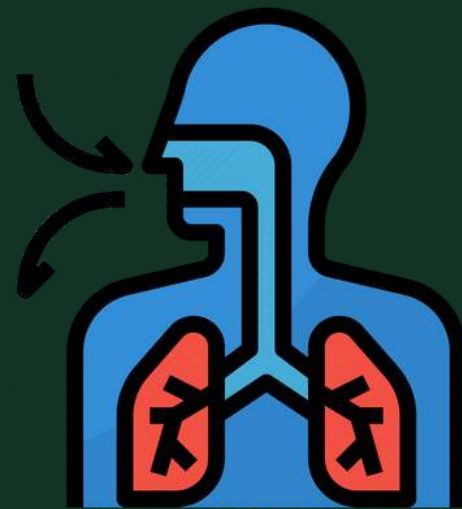


The basic and essential activities performed by a living organism to sustain and maintain life.

Nutrition



Respiration



Transportation



Excretion



Life Processes: Eat, Breathe, Pee, Repeat—just like a normal day.



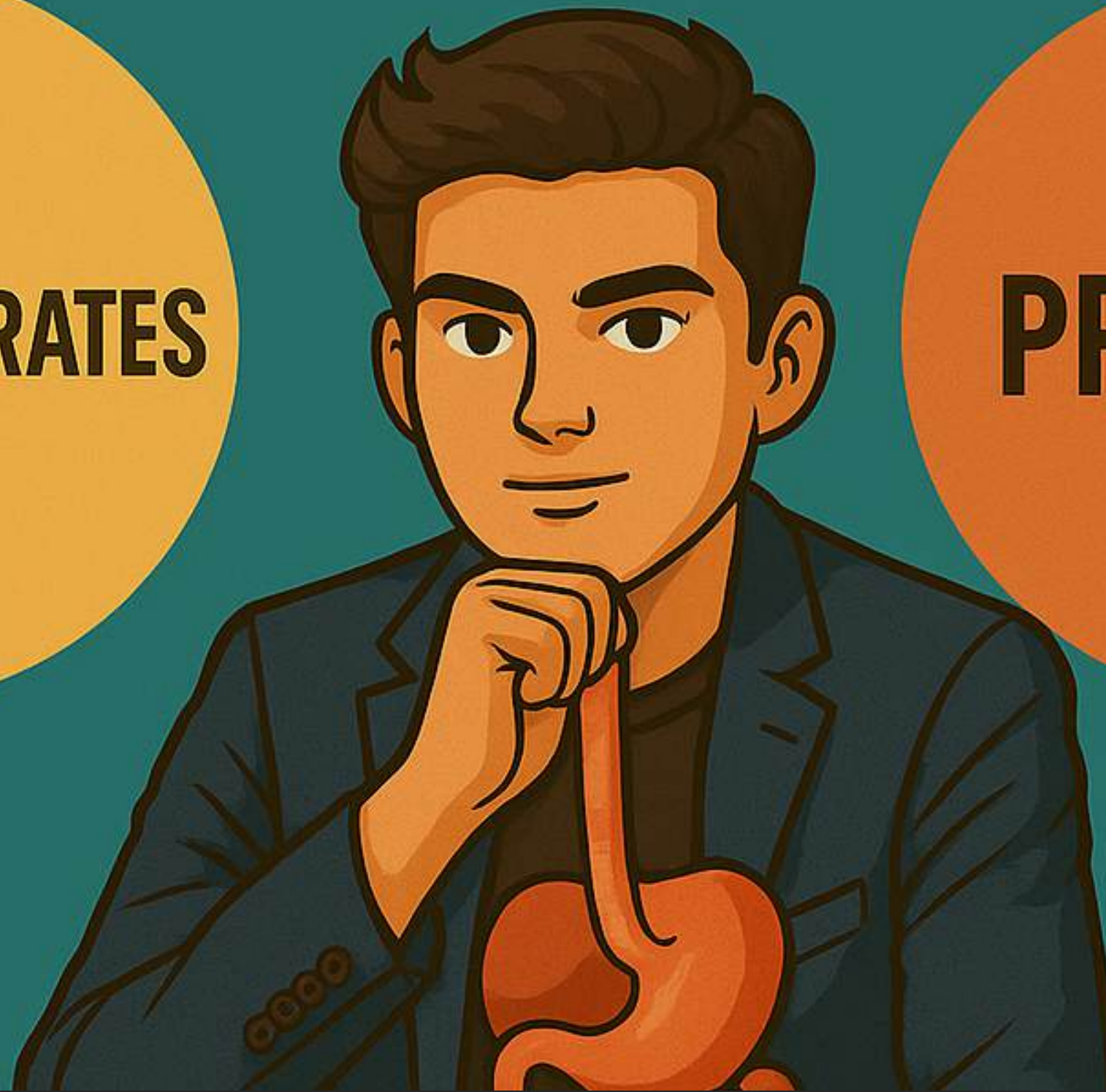
CARBOHYDRATES

PROTEINS

FATS

**VITAMIN &
MINERALS**

NUTRITION



NUTRITION



The process by which a living organism obtain and utilise food is called *Nutrition*.

Food gives us Energy, growth and development.

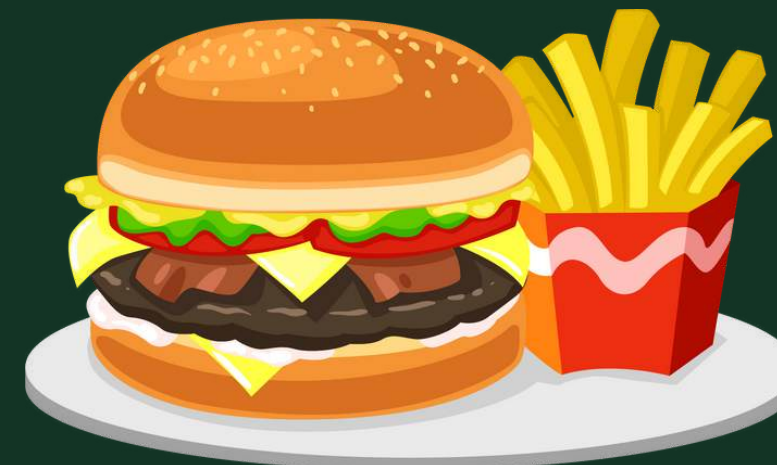
COMPONENTS OF FOOD



Carbohydrates



Proteins



Fats



Vitamin & Minerals

TYPES OF NUTRITION

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graph TD; A[TYPES OF NUTRITION] --> B[Autotrophic]; A --> C[Heterotrophic]; B --> D[Chemosynthesis]; B --> E[Photosynthesis];
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Autotrophic

It is a mode of nutrition in which organisms prepare their own food using simple inorganic substances like carbon dioxide and water in the presence of sunlight and chlorophyll. E.g., Green plants, algae, and blue-green algae

Chemosynthesis

A type of autotrophic nutrition in which organisms prepares their own food by utilizing chemical energy. E.g., Purple sulfur bacteria

Heterotrophic

It is a mode of nutrition in which organisms depend on other organisms (plants or animals) for their food because they cannot prepare their own food. e.g. All animals

Photosynthesis

A type of autotrophic nutrition in which organisms prepares their own food by utilizing light energy. E.g., Green plants, Cyanobacteria, Algae, Euglena

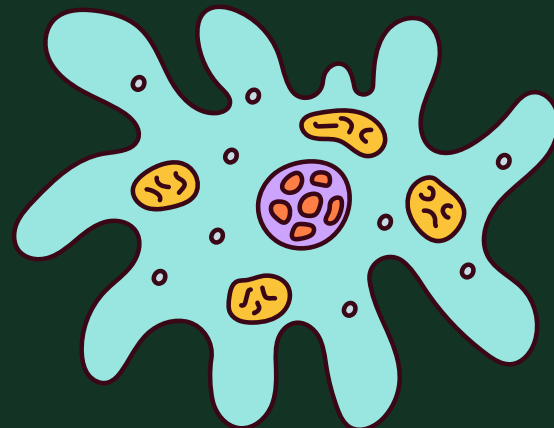
HETEROTROPHIC NUTRITION

Type of nutrition in which organisms do not possess the ability to synthesize their own food. They depend on autotrophs for their food supply, either directly or indirectly.

Holozoic

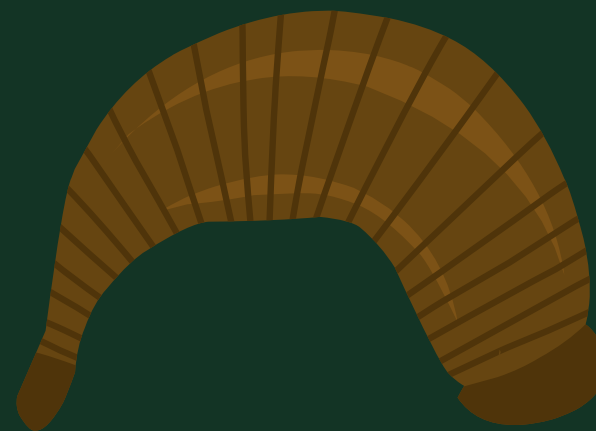
Animals take in solid food, which is broken down inside the body. ***E.g., Amoeba,***

animals



Saprophytic

Organisms feed on dead and decaying matter. ***E.g., Fungi***



Parasite

Parasites live inside or outside another organism (host) and derive nutrition from it. ***E.g. Cuscuta, Ticks, leech, etc.***



2017, 2020, 2021

Q. Which of the following organisms exhibits saprotrophic nutrition?

- a) Mango tree**
- b) Amoeba**
- c) Mushroom**
- d) Paramecium**

**CBSE Board Exams – 2019 (Set 1), 2021 (Set 3), 2023 (Set 2), 2025
(Sample Paper)**

Q. Which of the following organisms shows parasitic mode of nutrition?

- a) Amoeba**
- b) Algae**
- c) Mushroom**
- d) Cuscuta**

CBSE Sample Paper 2023, Practice Paper 2024 , Board Exam 2025 (Set 2, MCQ Section)

Q: An organism was discovered that lacks chlorophyll and survives by penetrating the tissues of other living organisms, from which it absorbs nutrients. However, it neither kills the host nor decomposes it. It also shows specificity by infecting only a particular plant species. Which of the following best describes the mode of nutrition in this organism?

- a) Saprotrophic nutrition**
- b) Holozoic nutrition**
- c) Parasitic autotrophic nutrition**
- d) Obligate parasitic heterotrophic nutrition**

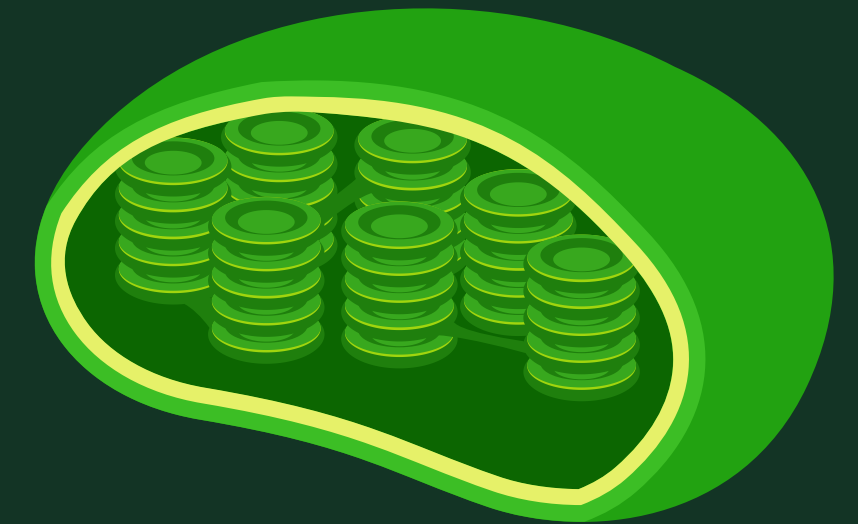
Photosynthesis



Raw material for photosynthesis: Sunlight, Chlorophyll, CO_2 and Water

SITE OF PHOTOSYNTHESIS

Chloroplasts are the site of photosynthesis which are green pigmented cell organelles containing **chlorophyll**.



MAIN EVENTS OF PHOTOSYNTHESIS

- Absorption of light energy by chlorophyll.
- Conversion of light energy into chemical energy + splitting (breaking) of water into hydrogen and oxygen.
- Reduction of CO_2 to carbohydrates.

Note: Plants store glucose in the form of “**starch**” while Humans store glucose in the form of “**glucose**”.

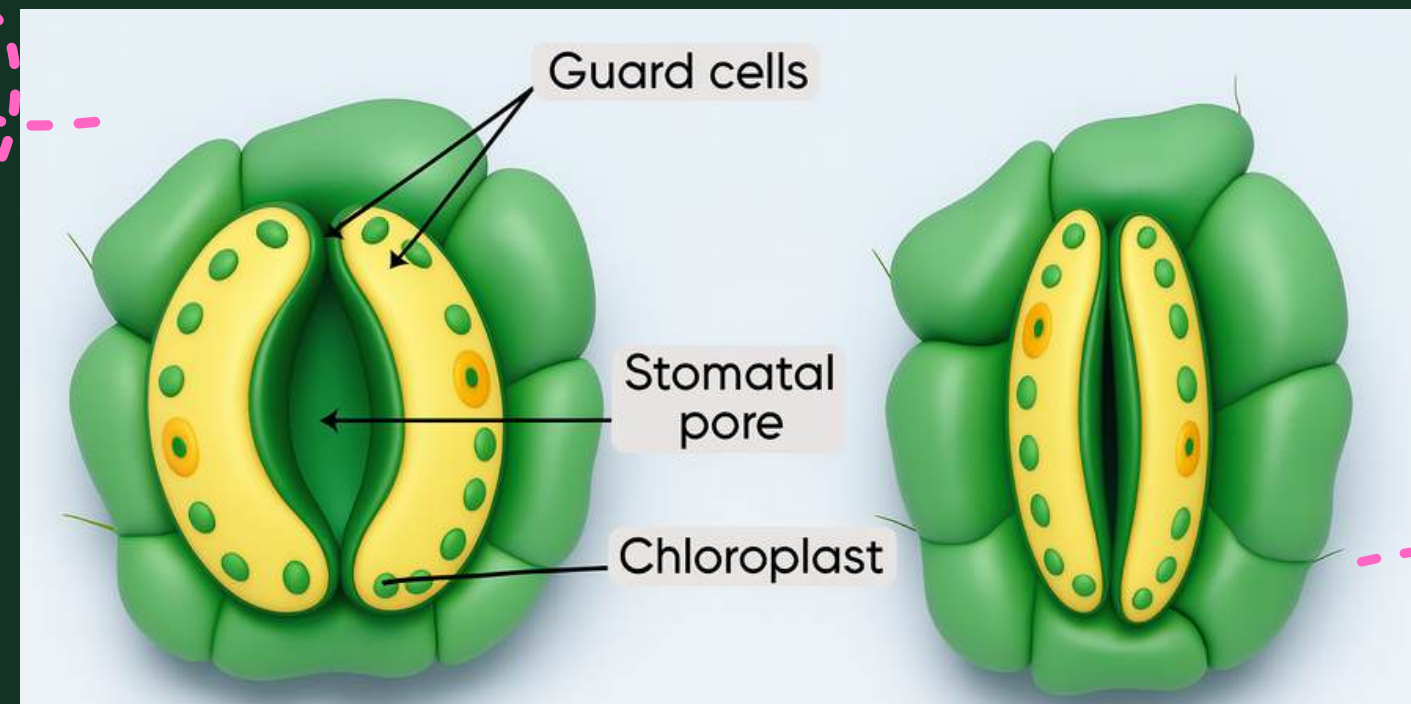
STOMATA AND ITS FUNCTIONS

Stomata are the tiny pores present on the surface of the leaves.

Functions of Stomata:

- Exchange of gases O_2/CO_2 .
- Loses large amount of water (water vapour) during transpiration.

It opens when water enters i.e., on swelling of guard cells.



It closes when water leaves i.e., on shrinking of guard cells.

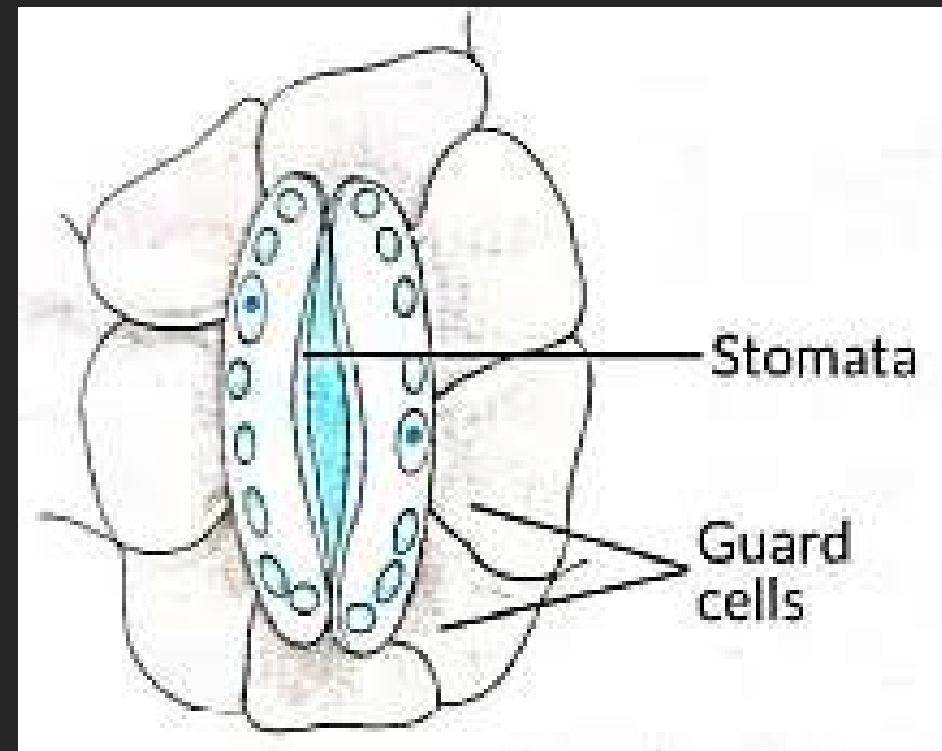
PHOTOSYNTHESIS IN DESERT PLANTS:

- Stomata open at night – to take in CO_2 (to reduce water loss).
- Stomata closed during the day – to conserve water.
- CO_2 taken at night is used for photosynthesis during the day.



2019, 2021, 2024 (Practice Paper)

Q. Which one of the following conditions is true for the state of stomata of a green leaf shown in the given diagram?



- (a) Large amount of water flows into the guard cells.**
- (b) Gaseous exchange is occurring in large amount.**
- (c) Large amount of water flows out from the guard cells.**
- (d) Large amount of sugar collects in the guard cells.**

2018, 2020 ,2022 , 2024 Sample Paper

Q.What are stomata? Where are they located in a plant? State two functions of stomata. Also explain how their opening and closing is regulated.

Ans. Stomata are tiny pores present mostly on the underside of leaves.

Functions (any two):

- *Exchange of gases (CO_2 in, O_2 out) during photosynthesis.*
- *Transpiration – Loss of excess water as vapour.*

Regulation: Controlled by guard cells.

- *When guard cells absorb water $\rightarrow$ become turgid $\rightarrow$ stomata open.*
- *When guard cells lose water $\rightarrow$ become flaccid $\rightarrow$ stomata close.*

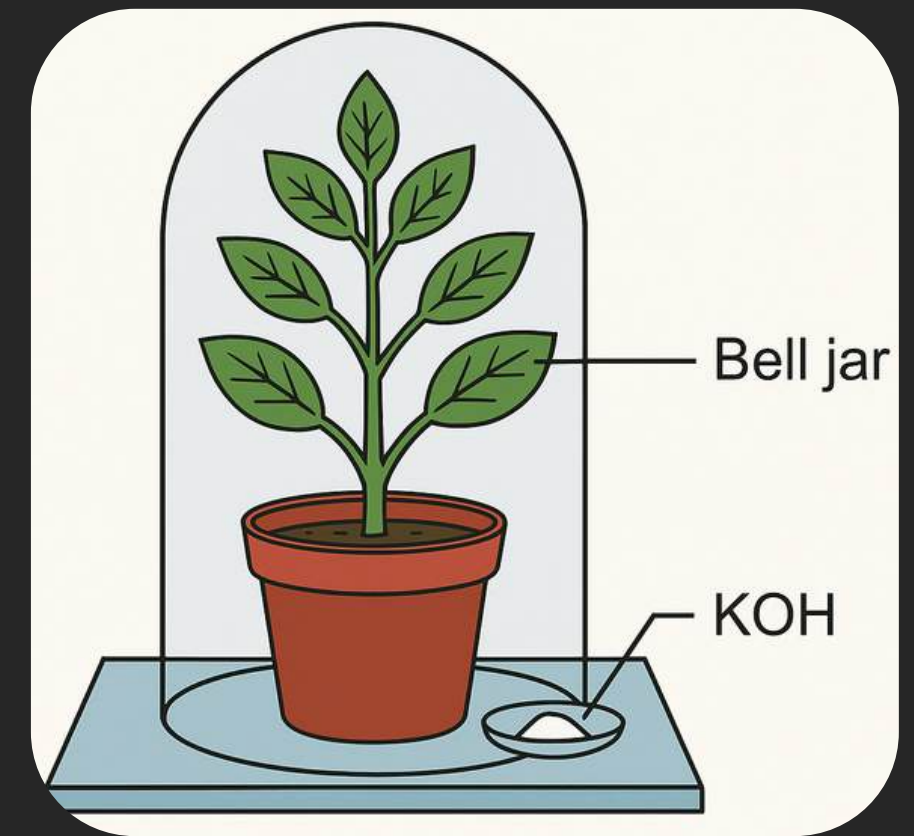
2020, 2022, 2023, 2024

Q. A student sets up an experiment to show that carbon dioxide is necessary for photosynthesis. He places a potted plant in a transparent air-tight bell jar. Inside the jar, he also keeps a watch glass containing potassium hydroxide (KOH).

(a) What is the role of KOH in this experiment?

(b) Why is the plant kept in sunlight after setting up the apparatus?

(c) What result will confirm that CO_2 is necessary for photosynthesis?

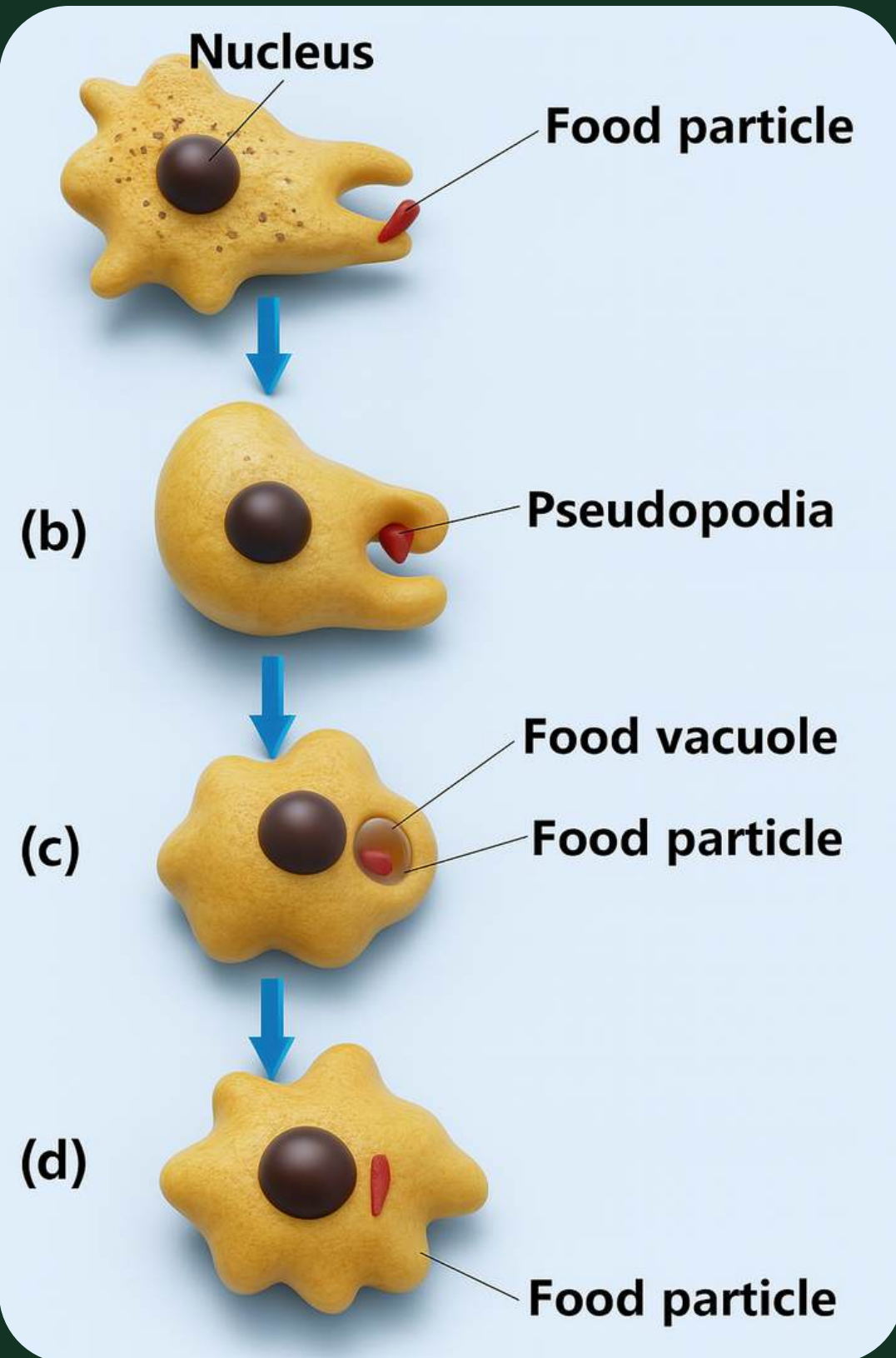


Ans. (a) KOH absorbs carbon dioxide from the air inside the bell jar, creating a CO_2 -free environment.

(b) To ensure that all other conditions for photosynthesis (light, chlorophyll, water) are available except CO_2 .

(c) After a few hours, if we test the leaves with iodine solution, they do not turn blue-black, which indicates that starch is not formed. → This confirms that CO_2 is essential for photosynthesis.

NUTRITION IN AMOEBA



1. Ingestion: Amoeba surrounds the food particle using finger-like projections called pseudopodia and forms a food vacuole.

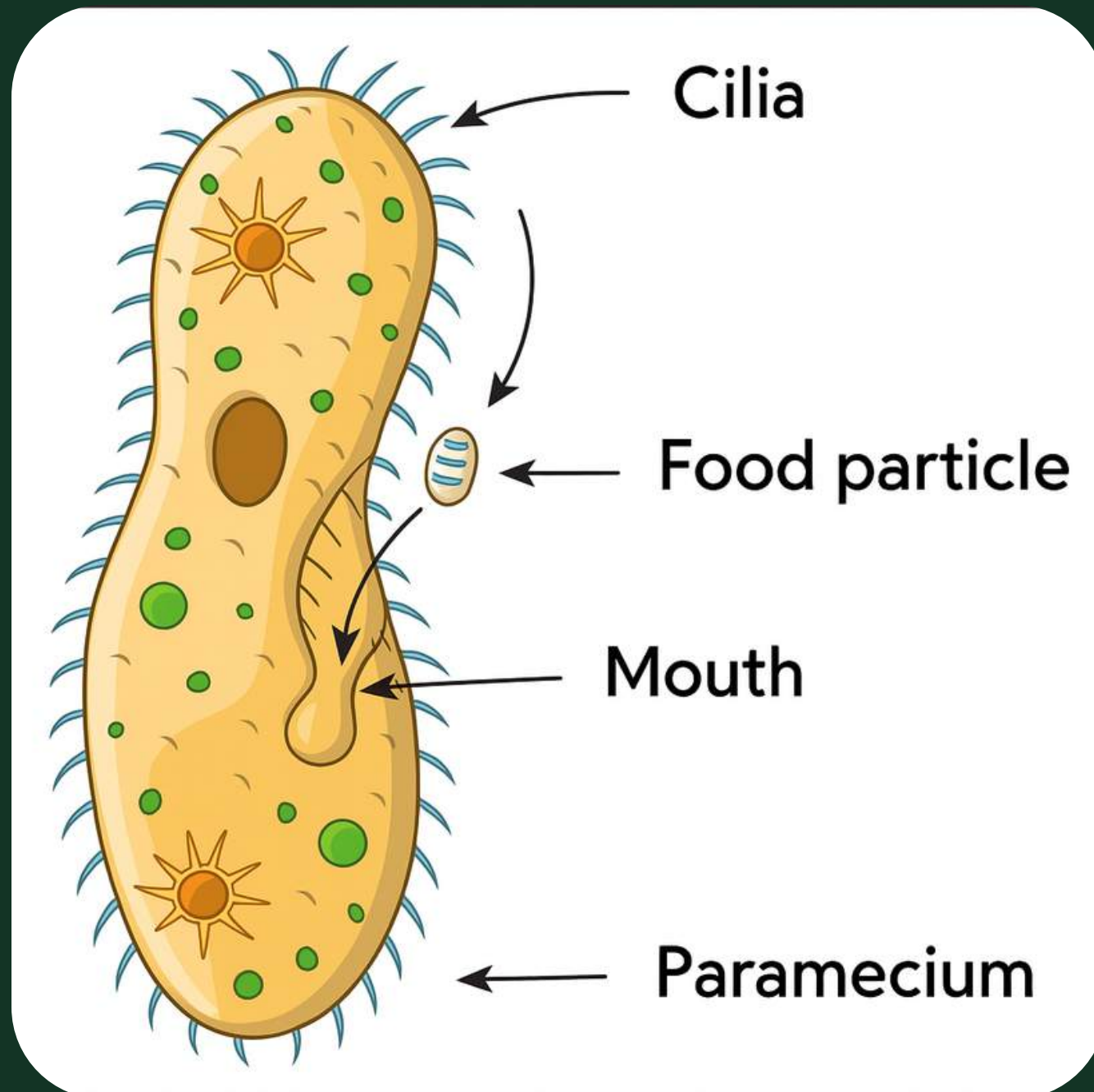
2. Digestion: Enzymes are secreted into the food vacuole, breaking down the food into simpler, soluble substances.

3. Absorption: The digested food is absorbed into the cytoplasm and used for energy, growth, and repair.

4. Assimilation: The absorbed nutrients are utilized in various life processes.

5. Egestion: The undigested food is expelled out of the body by rupturing the cell membrane at any point.

NUTRITION IN PARAMECIUM



Paramecium also follows holozoic nutrition, but its method of ingestion differs due to its structure:

- 1. Ingestion:** Paramecium uses cilia (hair-like structures) to sweep food particles like bacteria through the oral groove into the gullet where a food vacuole is formed.
- 2. Other steps (Digestion, Absorption, Assimilation, Egestion)** are similar to Amoeba and occur inside the food vacuole.

Cilia (present all over the body)



Take in food



At a specific spot

2019, 2020, 2022, 2024

Q. Differentiate between Amoeba and Paramecium based on:

1. Mode of nutrition

2. Locomotion (movement)

<i>Feature</i>	<i>Amoeba</i>	<i>Paramecium</i>
<i>Mode of Nutrition</i>	<i>Holozoic; ingests food using pseudopodia</i>	<i>Holozoic; uses cilia to sweep food into oral groove</i>
<i>Locomotion</i>	<i>Moves by forming pseudopodia (false feet)</i>	<i>Moves with the help of cilia (hair-like structures)</i>

NUTRITION IN HUMANS

Ingestion: Intake of food material



Digestion: Breakdown of complex food



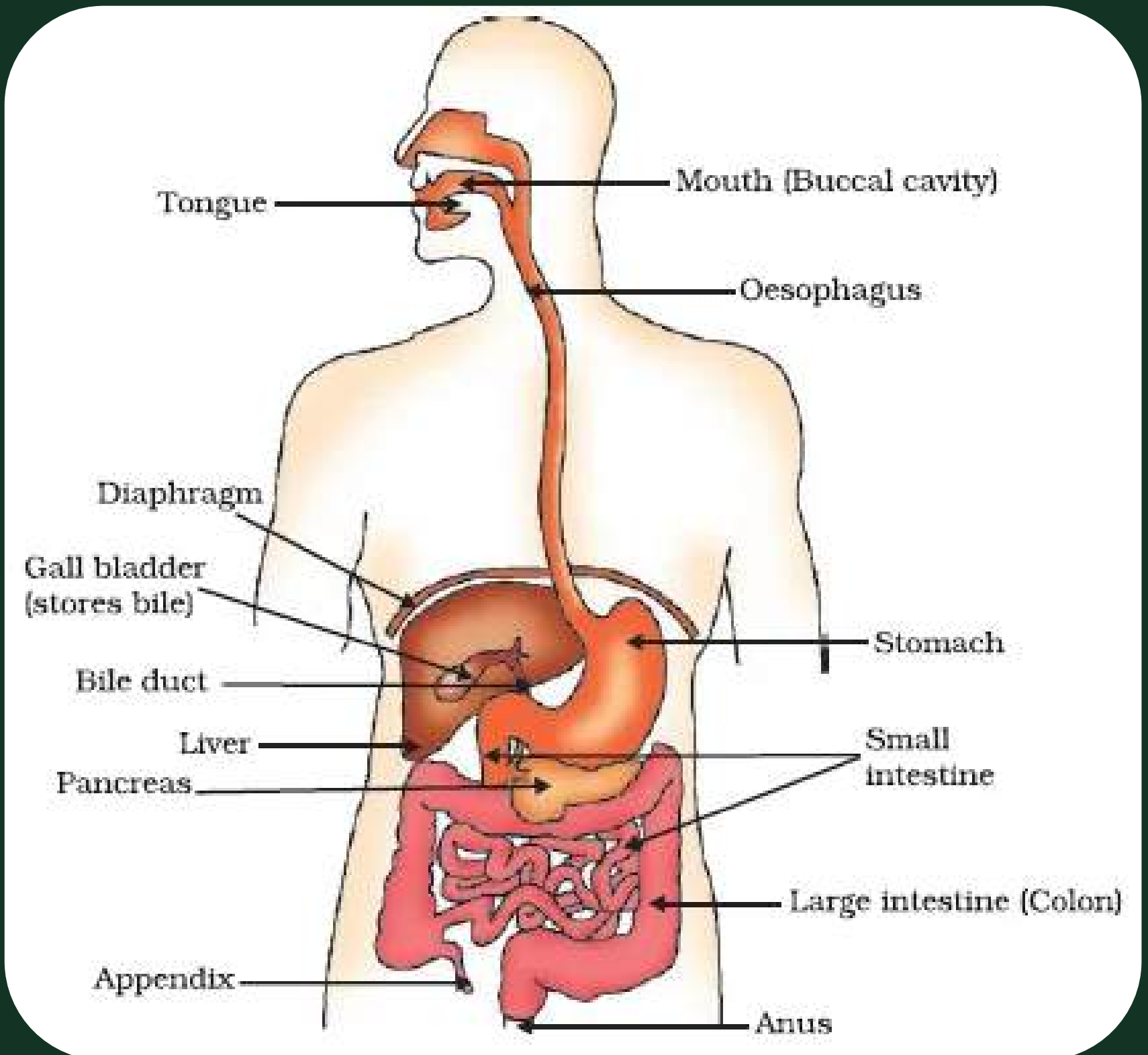
Absorption: Movement of digested food



Assimilation: Utilization of food products



Egestion: Removal of waste



NUTRITION IN HUMANS

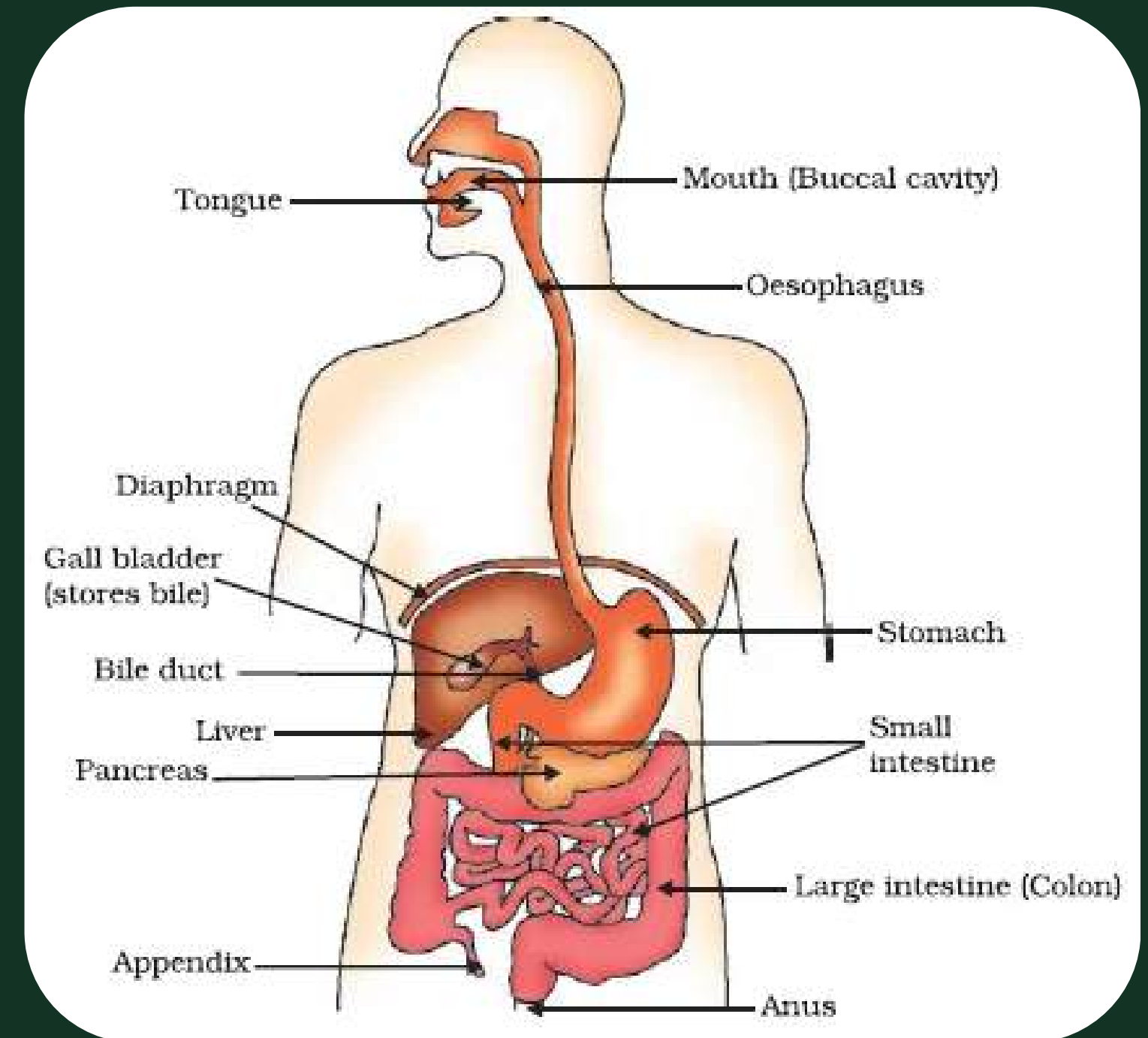
1. Alimentary Canal: A long muscular tube from mouth to anus.

2. Teeth crush food → forms uniform texture.

- Salivary glands secrete saliva.
- Contains enzyme **salivary amylase** → breaks down starch into sugars.
- Tongue mixes food with saliva and helps in swallowing.

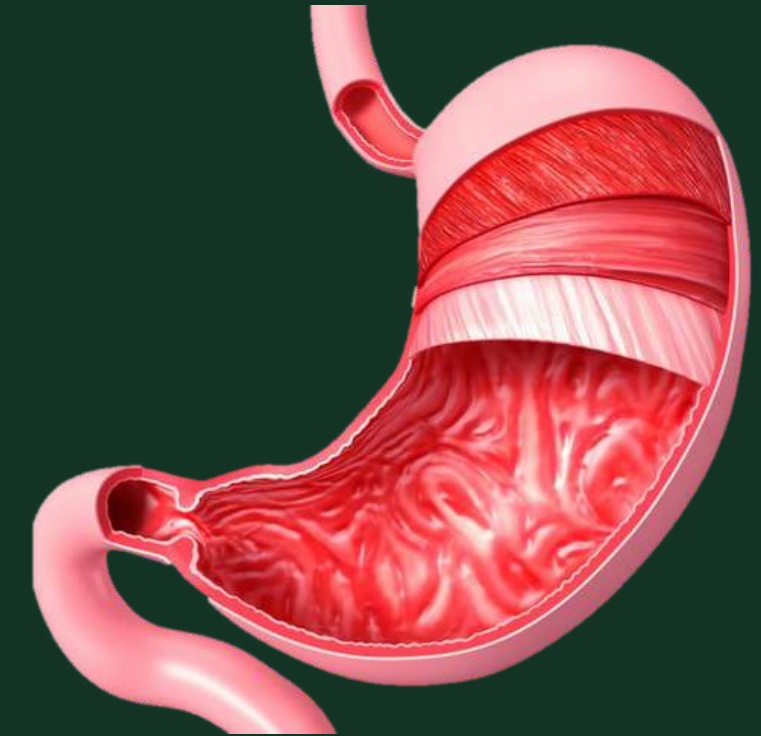
3. Oesophagus (Food Pipe)

- Connects mouth to stomach. No digestion occurs here.
- Food is pushed down through **peristalsis** – rhythmic contraction of muscles in the wall of the alimentary canal.



5. Stomach: Large muscular organ.

- Secretes gastric juice containing:
 - a. **HCl** – creates acidic medium (helps pepsin action, kills germs).
 - b. **Pepsin** – digests proteins.
 - c. **Mucus** – protects inner lining of stomach.
- Food stays in stomach for 3–4 hours.



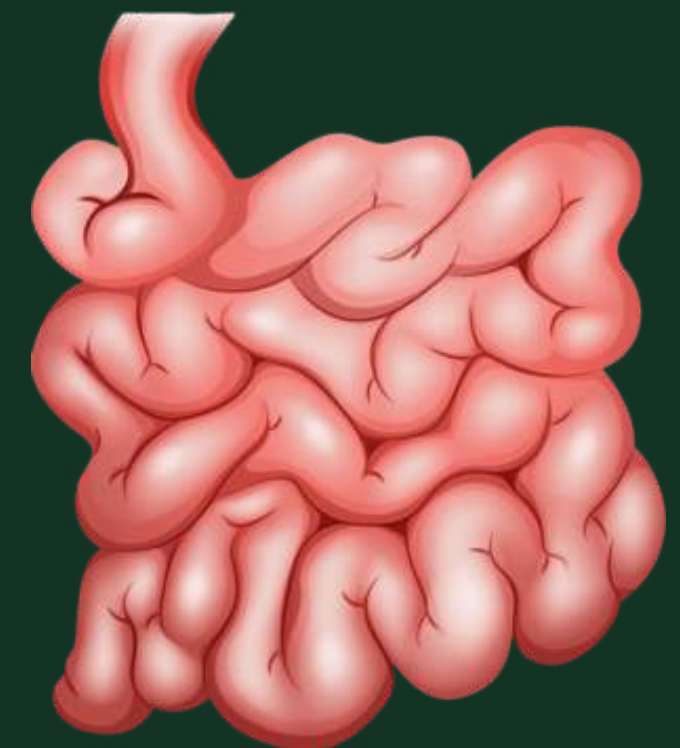
6. Small Intestine: Longest part (~7 meters in humans), coiled. Site of complete digestion and absorption.

Receives secretions from:

- **Liver (Bile juice)**: Makes food alkaline and Emulsifies fats (breaks into small droplets)
- **Pancreas (Pancreatic juice)**: Trypsin (proteins)
 - Lipase (fats)
 - *Pancreatic amylase* – helps in digestion of carbohydrates
- **Intestinal glands**: Secrete intestinal juice:

Final breakdown of: **Proteins → Amino acids**

- **Carbohydrates → Glucose**
- **Fats → Fatty acids + Glycerol**



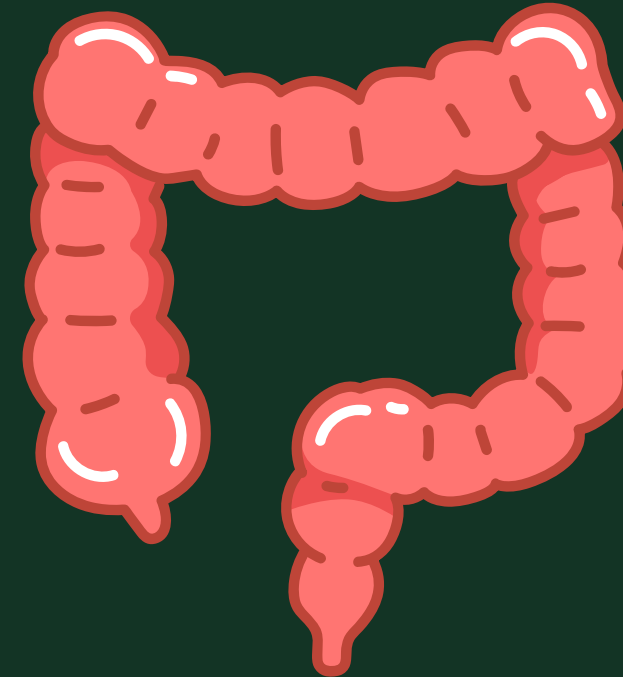
Absorption of Food in Small Intestine

- Villi are finger-like projections in the inner wall of the small intestine.
- They increase surface area for absorption.
- Rich blood supply helps transport absorbed nutrients to all body cells.
- Nutrients are used for energy, growth, and repair of tissues.



7. Large Intestine

- Absorbs water from undigested food.
- Prepares waste for excretion.



8. Anus

- Egestion: Removal of undigested waste.
- Controlled by anal sphincter.

Herbivores have a longer digestive system because plant food contains hard-to-digest cellulose, which requires more time for proper breakdown and absorption.



2019, 2021, 2023, 2025

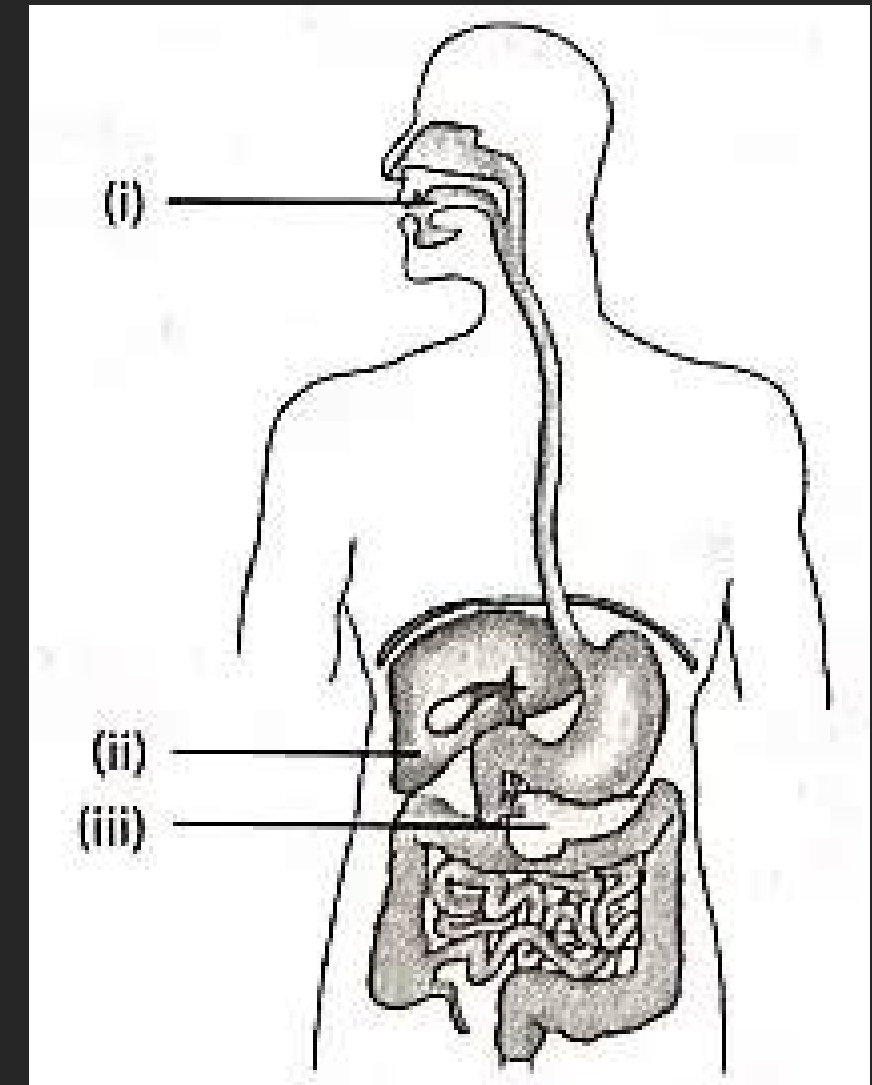
Q. Which enzyme is secreted in the mouth and what does it digest?

- a) Pepsin – proteins**
- b) Amylase – starch**
- c) Lipase – fats**
- d) Trypsin – proteins**

2022 CBSE Board Exam (Set 2 – MCQ Section)

Q. In human alimentary canal, the specific enzyme/ juice secreted in locations (i), (ii) and (iii) are

- (a) (i) Amylase (ii) Pepsin (iii) Bile**
- (b) (i) Amylase (ii) Bile (iii) Trypsin**
- (c) (i) Lipase (ii) Amylase (iii) Pepsin**
- (d) (i) Trypsin (ii) Bile (iii) Amylase**



2020, 2022, 2024

Q. In human beings, when the process of digestion is completed, the (i) proteins, (ii) carbohydrates, and (iii) fats are respectively finally converted into:

- (a) (i) Amino acids, (ii) glucose and (iii) fatty acids**
- (b) (i) Amino acids, (ii) glucose, (iii) fatty acids and glycerol**
- (c) (i) Glucose, (ii) fatty acids and glycerol, (iii) amino acids**
- (d) (i) Sugars, (ii) amino acids, (iii) fatty acids and glycerol**

2018, 2020, 2022, 2024 (Sample Paper)

Q. State the role of the following in human digestive system:

(a) Hydrochloric acid

(b) Bile

(c) Pepsin

Ans. (a) Hydrochloric acid (HCl): Creates an acidic medium in the stomach.

- Helps activate the enzyme pepsin.*
- Kills harmful microbes in food.*

(b) Bile :Secreted by the liver, stored in the gall bladder.

- Helps in emulsification of fats (breaks down large fat globules into smaller ones).*
- Makes the medium alkaline so that pancreatic enzymes can act.*

(c) Pepsin :Protein-digesting enzyme present in the stomach.

- Breaks down proteins into simpler peptides in an acidic medium.*

2019, 2020, 2021, 2022, 2024

Q. What are villi? Where are they located in the human body? How do they help in absorption of food?

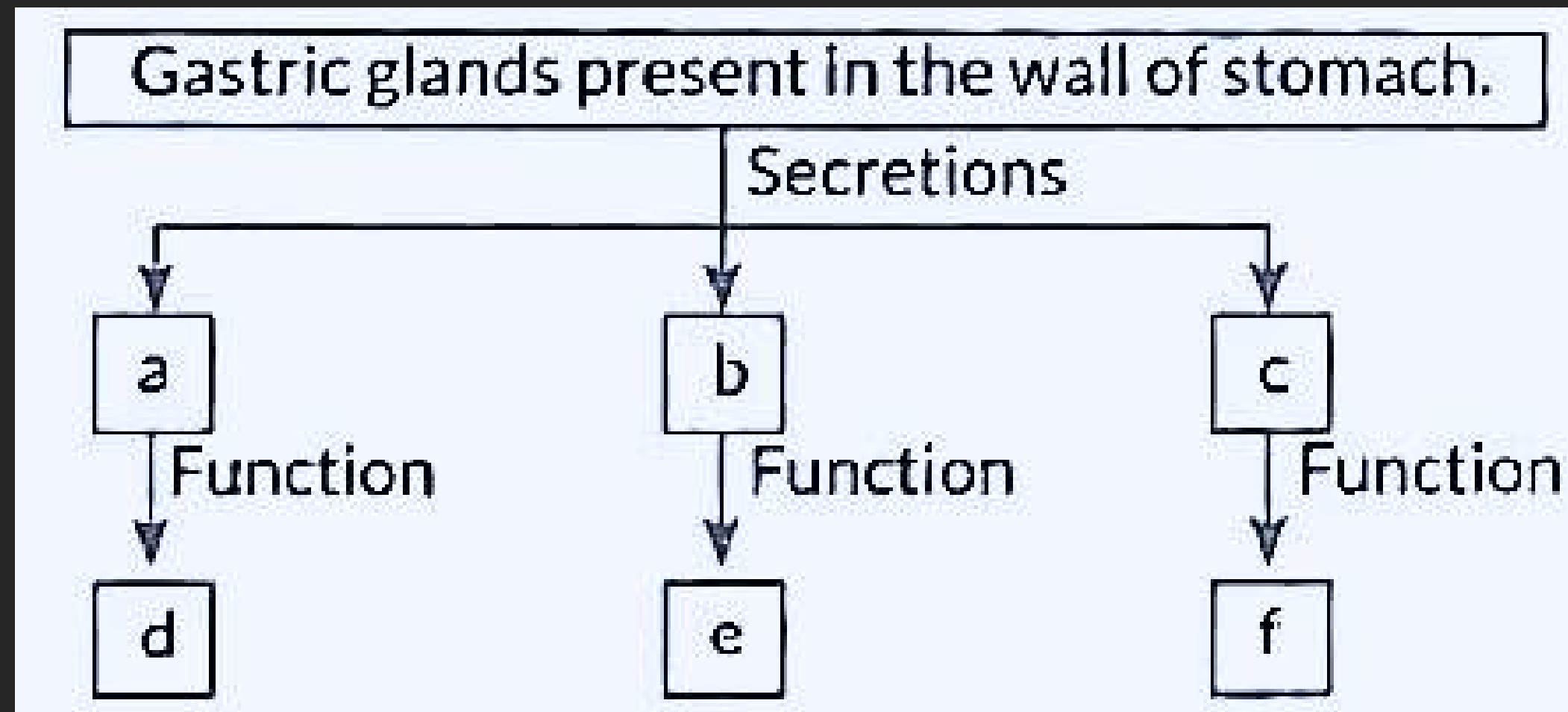
Ans. Villi are finger-like projections present on the inner lining of the small intestine.

They help in absorption of food because:

- They increase the surface area for absorption.*
- They have thin walls and are rich in blood capillaries, which help in the quick transfer of digested food into the bloodstream.*

2014,2020

Q. Complete the following flow chart as per the given instructions.





RESPIRATION

BREATHING

It is exchange of gases (Oxygen and carbon dioxide).

Inhalation: Taking air inside the body.

Exhalation: Releasing air outside the body.

RESPIRATION:

Process by which organisms break down glucose in their cells to release energy in the form of ATP, which is used for various life processes.



BREATHING	RESPIRATION
Physical process	Biochemical process
No enzymes involved	Large number of enzymes are involved
Confined to certain organs	Occurs in all the cells of the body
No energy is released; rather, energy is used	Energy is released in the form of ATP
Extracellular process	Intracellular process
Intake of fresh air and removal of foul air	Oxidation of food to form carbon dioxide, water, and energy

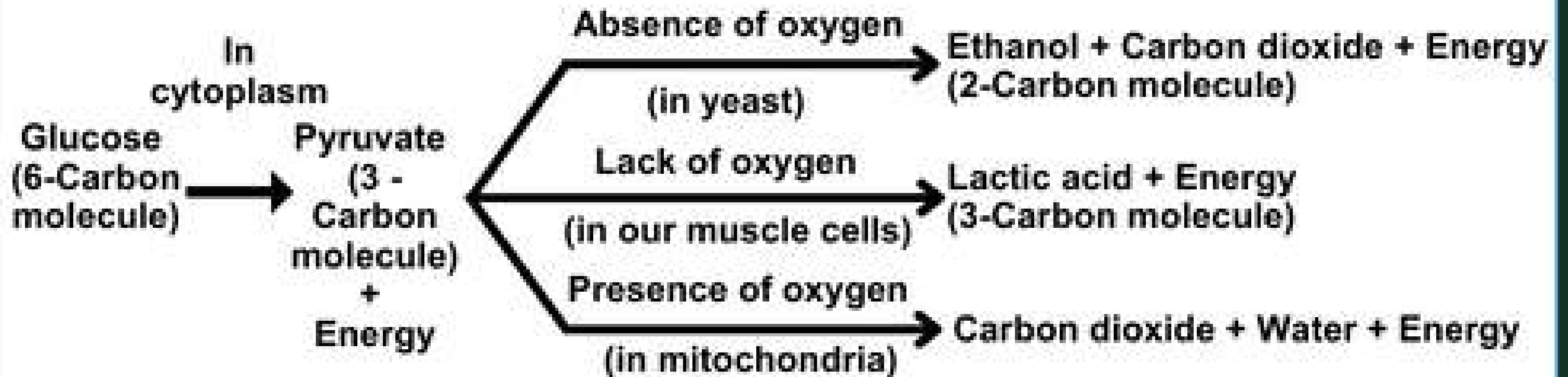
It is the process by which *food is oxidized to release energy*. For this *oxygen is required from outside of the body to breakdown the food*.



TYPE OF RESPIRATION

Aerobic Respiration	Anaerobic Respiration
Takes place in the presence of oxygen	Takes place in the absence of oxygen
Glucose is completely broken down into carbon dioxide and water	Glucose is partially broken down into alcohol or lactic acid
More energy (ATP) is released	Less energy is released
Occurs in most plants and animals (including humans)	Occurs in some microorganisms (e.g., yeast) and muscle cells during lack of oxygen

BREAKDOWN OF GLUCOSE



TYPES OF ANAEROBIC RESPIRATION

i. Alcoholic fermentation

It is the process of incomplete breakdown of sugar into ethanol, CO₂ to release energy. This process occurs mainly in yeast which is used to produce beer, urine, etc.



ii. Lactic fermentation

It is process of incomplete breakdown of sugar into lactic acid + energy in some bacteria and in our muscle cells

During intense exercise, muscles rapidly convert glucose to lactic acid for quick energy, using up stored ATP. This lactic acid buildup causes fatigue and cramps.



2020, 2022, 2024

Q. Name the substances whose build up in the muscles during vigorous physical exercise may cause cramps?

- (a) Ethanol + Carbon dioxide + Energy**
- (b) Lactic acid + Energy**
- (c) Carbon dioxide + Water + Energy**
- (d) Pyruvate**

2016, 2019, 2022, 2024

Q. Which gas is released during anaerobic respiration in yeast?

- a) Oxygen**
- b) Carbon dioxide**
- c) Hydrogen**
- d) Nitrogen**

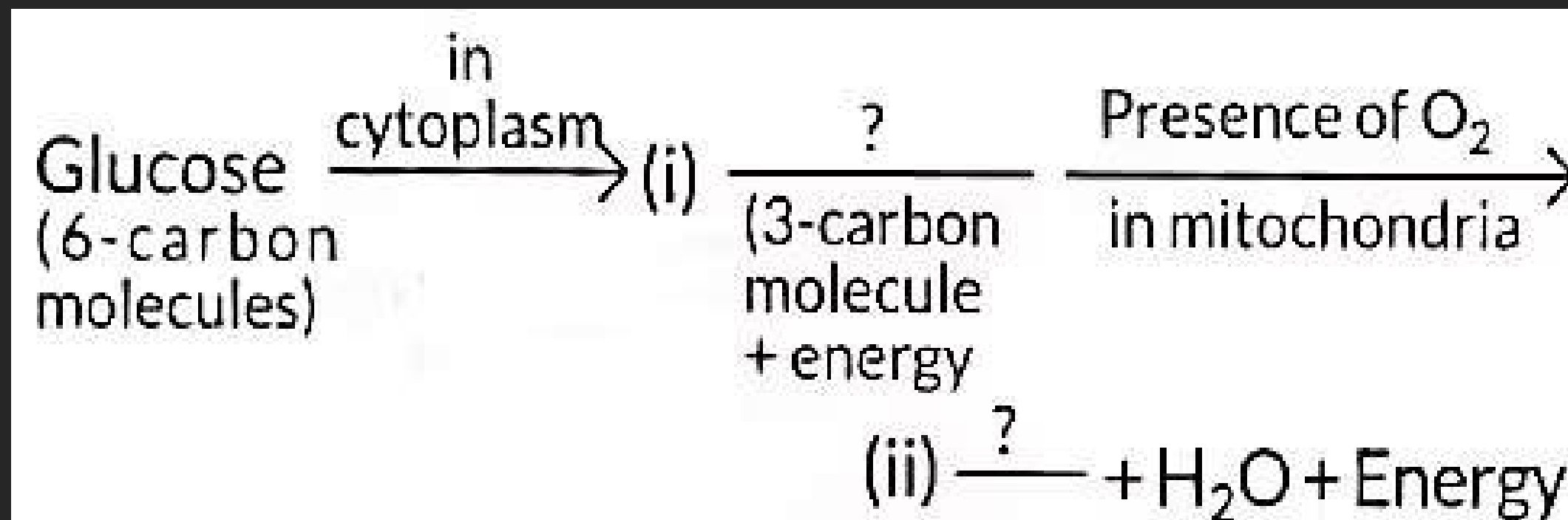
2021, 2023, 2025 (Sample Paper)

Q. A boy was running a 400-meter race. After finishing, he was breathing heavily. His friend noticed the flaring of nostrils and rapid chest movements. What causes an increase in the breathing rate after intense physical activity?

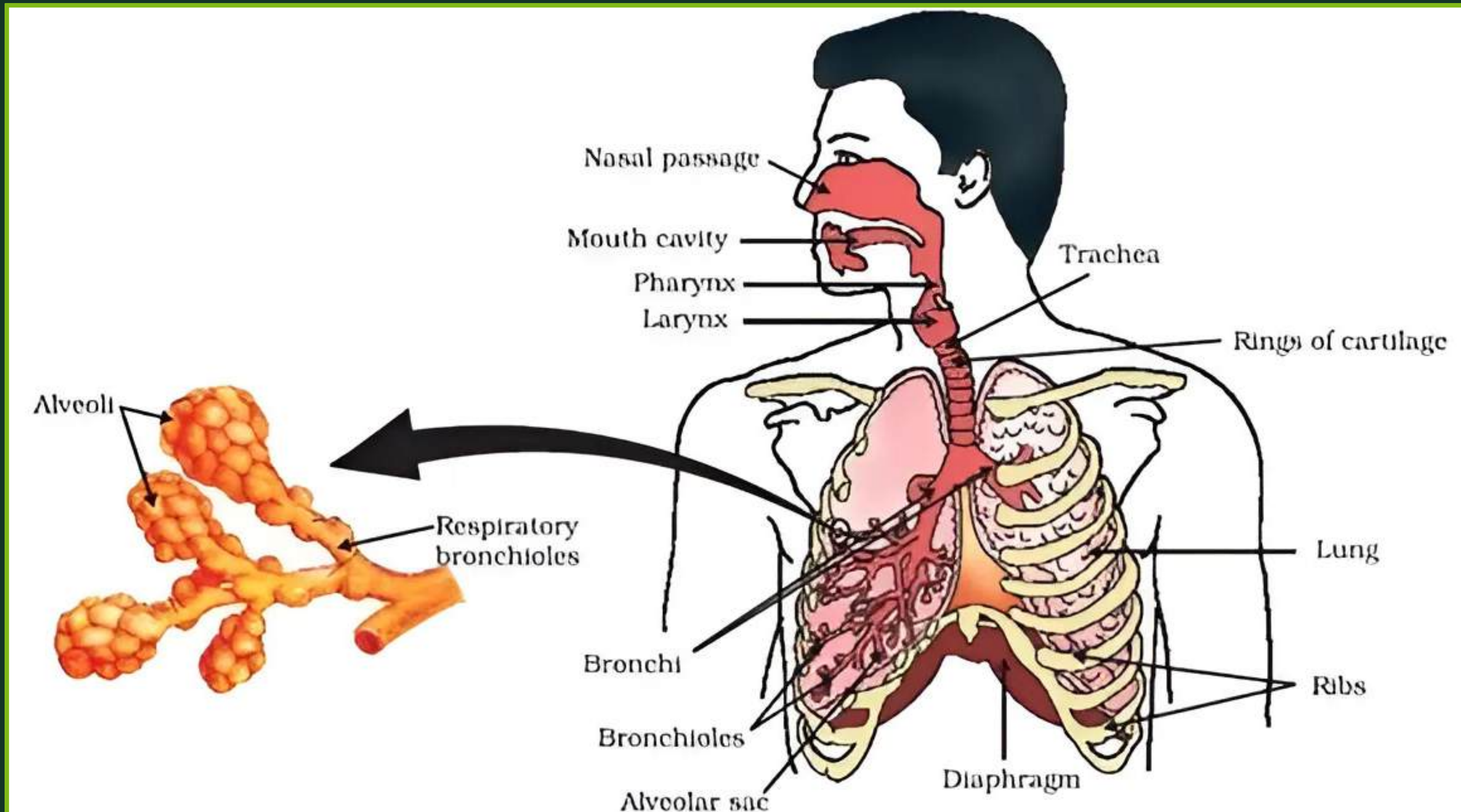
- a) Increased oxygen levels in blood**
- b) Decreased CO₂ levels in blood**
- c) Accumulation of lactic acid and increased CO₂ concentration**
- d) Decrease in diaphragm contraction**

2020, 2022, 2024

Q. Complete the following pathway showing the breakdown of glucose.

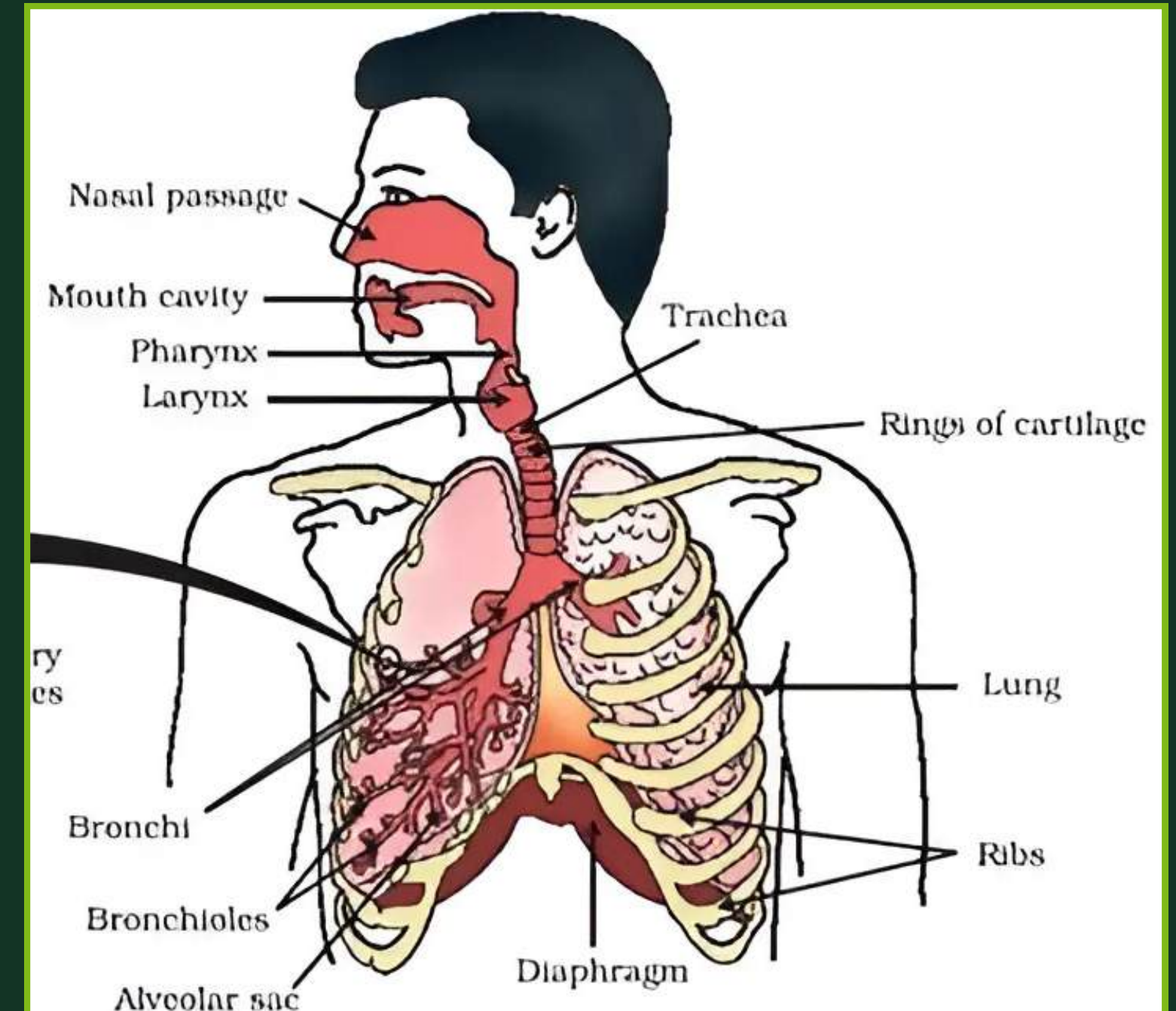


HUMAN RESPIRATORY SYSTEM

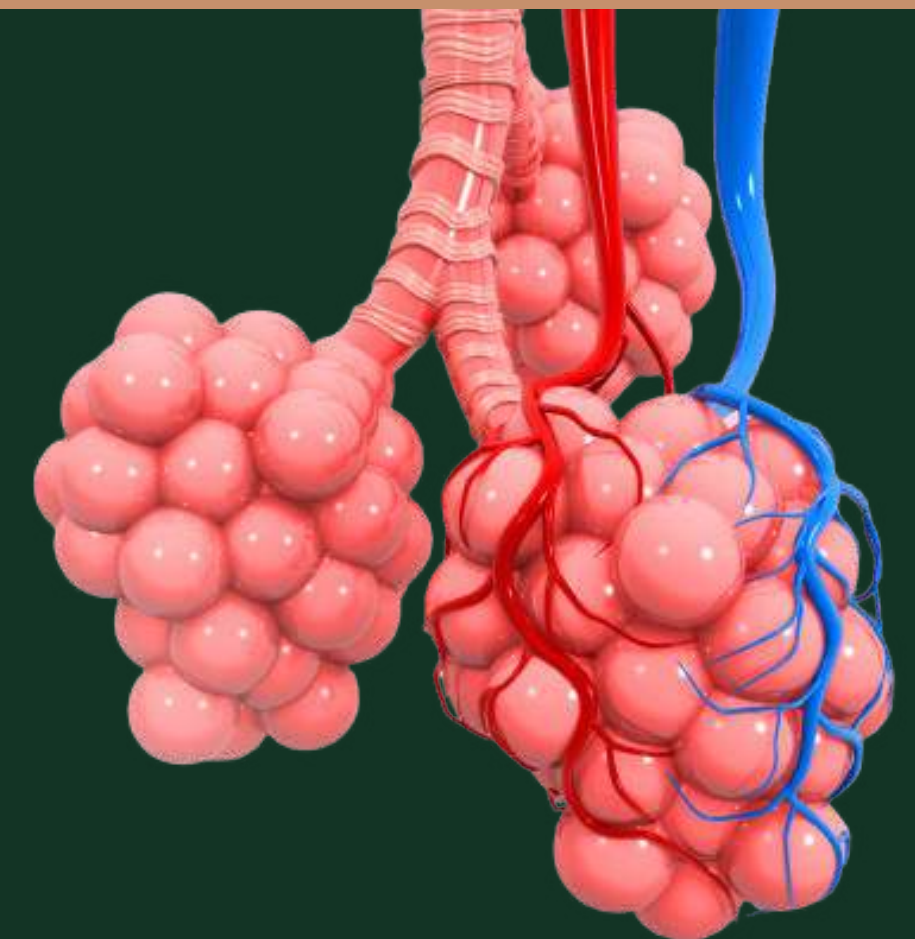
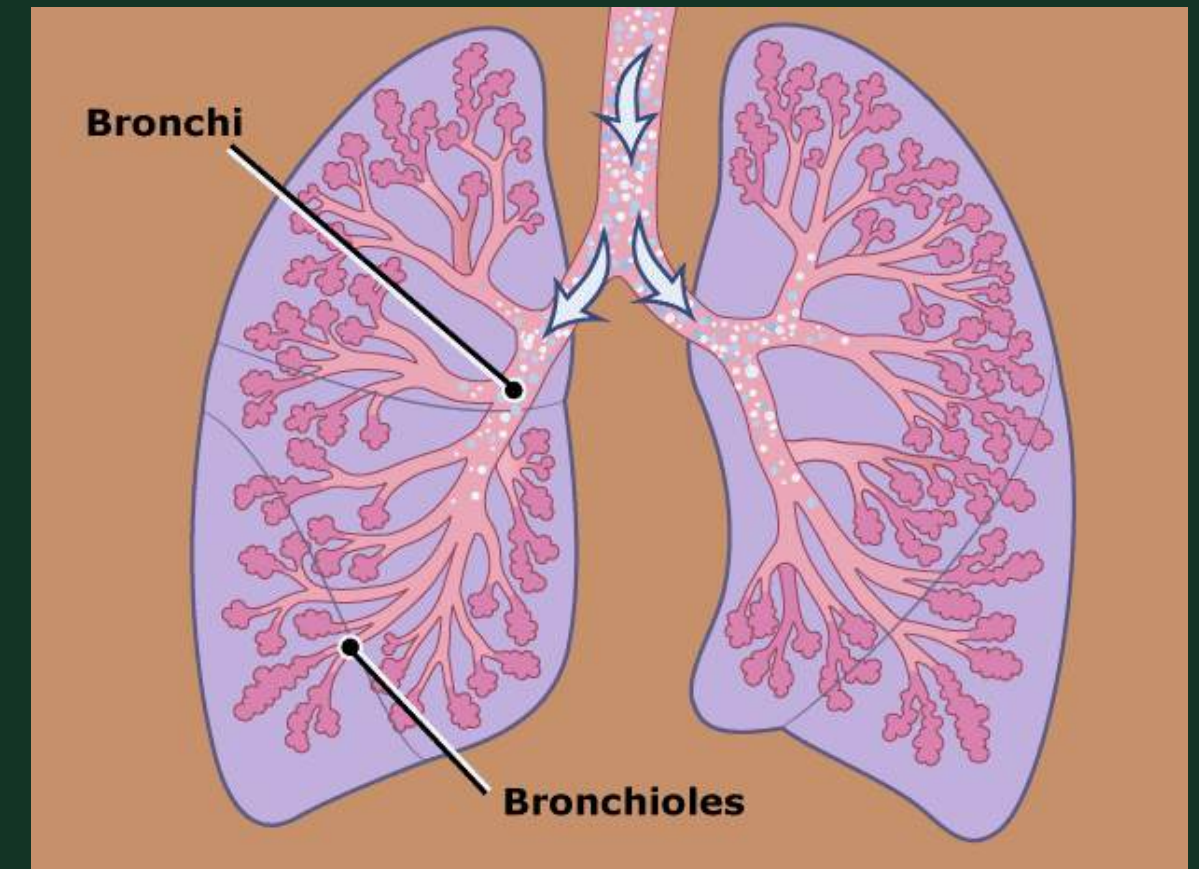


Passage of air through Respiratory System

- **Nostril:** Air is taken into the body. Filtered by fine hairs and mucus
- **Nasal Passage:** It is a channel for airflow through the nose. Warms, moistens, and further filters the air.
- **Pharynx:** Common passage for both air and food which ensure that the air-passage does not collapse.
- **Larynx:** Located at the upper part of the trachea. Contains vocal cords → produces sound. Also allows air to pass into the trachea.
- **Trachea:** Tube that connects larynx to bronchi. Lined with rings of cartilage to keep the airway open. Conducts air into the lungs.

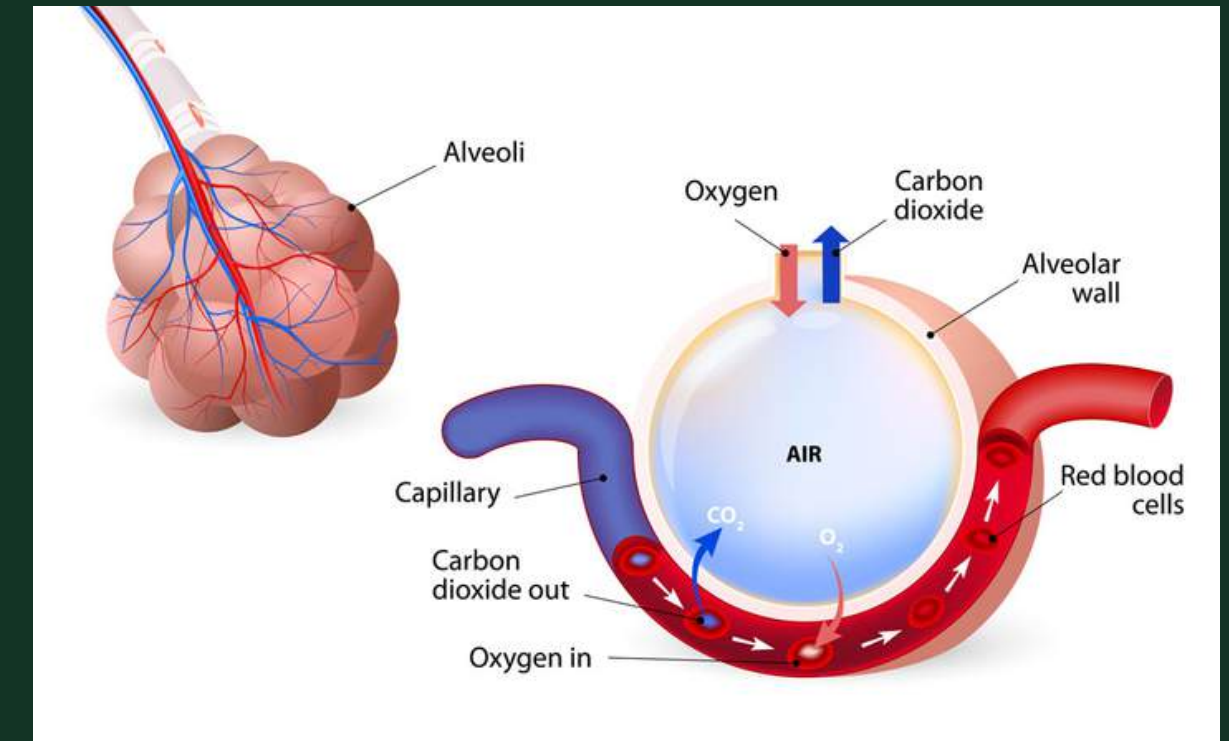


- **Bronchi:** Trachea splits into two bronchi (left and right), one for each lung. Carry air into the lungs.
- **Bronchioles :** The bronchi further divide into narrower tubes called bronchioles as they enter deeper into the lungs. Bronchioles are small air passages that lead directly to the alveoli. They do not have cartilage rings and can contract or expand to regulate airflow.
- **Alveoli (Singular: Alveolus) :**Alveoli are tiny, balloon-like air sacs located at the end of bronchioles. Thin-walled and moist for easy gas diffusion. Surrounded by a dense network of blood capillaries. **They are the site of gas exchange in the lungs.** Each lung contains millions of alveoli, providing a large surface area for efficient diffusion.
- **Blood Capillaries:** Tiny blood vessels surrounding each alveolus. Allow exchange of gases between blood and alveolar air. Also involved in transporting oxygen to body tissues and collecting CO₂ from them.



Alveolus gas exchange

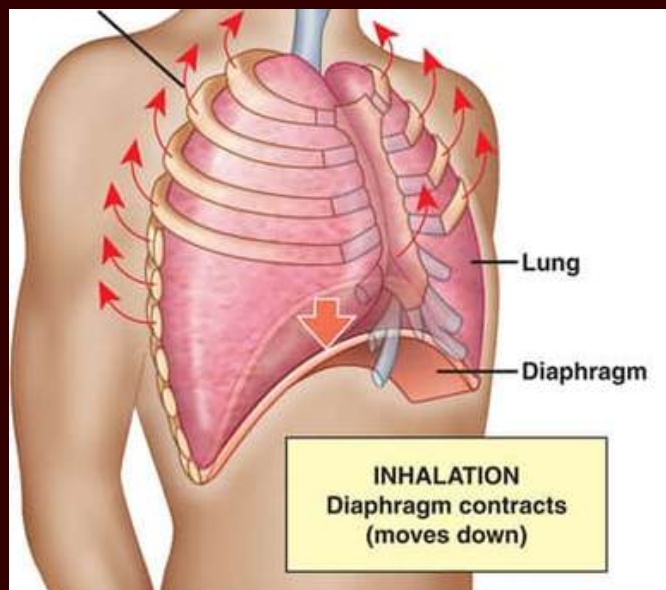
- Oxygen (O_2) from alveolar air diffuses into the blood capillaries.
- Carbon dioxide (CO_2) from blood diffuses into the alveolar space to be exhaled.
- This exchange occurs through diffusion due to the difference in concentration gradients.



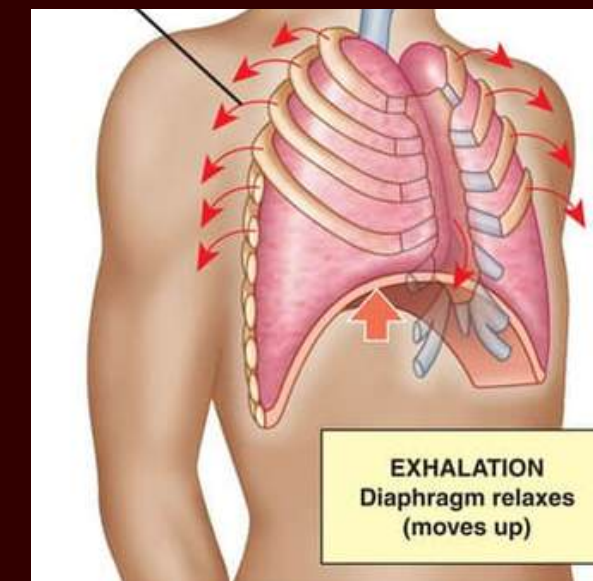
Hemoglobin: Iron-containing protein present in Red Blood Cells (RBCs) which is responsible for the red colour of blood

- Plays a key role in transporting oxygen from lungs to body tissues
- Affinity of Hemoglobin for various gases: $CO > O_2 > CO_2$

Breathing Mechanism (Inhalation and Exhalation)



Inhalation: Ribs move up,
diaphragm flattens
Chest cavity increases →
air is sucked into lungs



Exhalation: Ribs move down,
diaphragm relaxes (domes up)
Chest cavity decreases → air is
pushed out

2019, 2021, 2023, 2025 (Sample Paper)

Q. Give reasons for the following:

- (a) Rings of cartilage are present in the throat.**
- (b) Lungs always contain a residual volume of air.**
- (c) The diaphragm flattens and ribs are lifted up when we breathe in.**
- (d) Walls of alveoli contain an extensive network of blood vessels.**

Ans.

- (a) To keep the windpipe open and prevent its collapse, ensuring continuous airflow to the lungs.*
- (b) To allow continuous exchange of gases and prevent lung collapse.*
- (c) To increase chest cavity volume, creating low pressure so air can enter the lungs.*
- (d) To allow efficient exchange of oxygen and carbon dioxide between lungs and blood.*

2020, 2022, 2024

Q. Why is respiratory pigment needed in multicellular organisms with large body sizes?

Ans. In multicellular organisms with large body sizes, diffusion alone is not sufficient to supply oxygen to all body cells efficiently. Hence, a respiratory pigment like haemoglobin is required.

Haemoglobin, present in red blood cells (RBCs), has a high affinity for oxygen. It helps in absorbing oxygen from the lungs and transporting it to all body tissues. It also carries carbon dioxide from tissues back to the lungs for exhalation

2018, 2020, 2022, 2024 (Practice Paper)

Q. Describe the exchange of gases in alveoli and Why are alveoli richly supplied with blood vessels?

Ans. Alveoli are tiny balloon-like structures in the lungs where gaseous exchange occurs.

- Each alveolus is surrounded by a network of thin-walled blood capillaries.*
- Oxygen present in the inhaled air diffuses from the alveolar air space into the blood in the capillaries.*
- Carbon dioxide from deoxygenated blood diffuses into the alveoli to be exhaled.*
- This exchange occurs by simple diffusion due to concentration gradients.*

Enables efficient exchange of gases with the blood

RESPIRATION IN PLANTS

Gaseous exchange occur through :

(a) Stomata in leaves

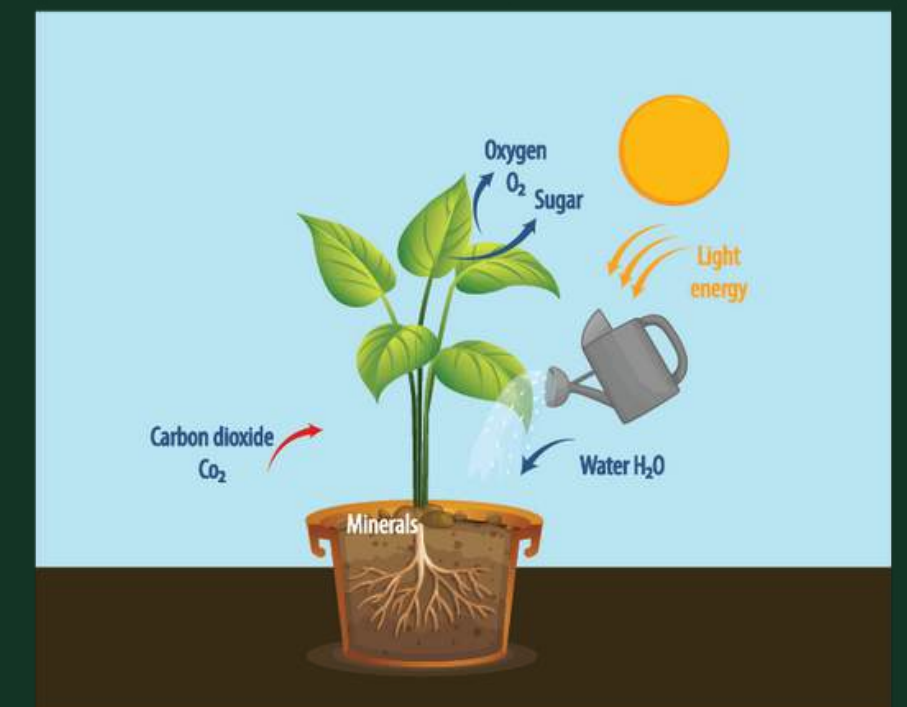
(b) Lenticles in stems

(c) General surface of Roots



At night, when photosynthesis stops, CO_2 elimination is the main activity.

During the day, CO_2 from respiration is used in photosynthesis, so no CO_2 is released—instead, oxygen is the primary gas released.



2019, 2021, 2023, 2025

Q. Which of the following statements about respiration in plants is correct?

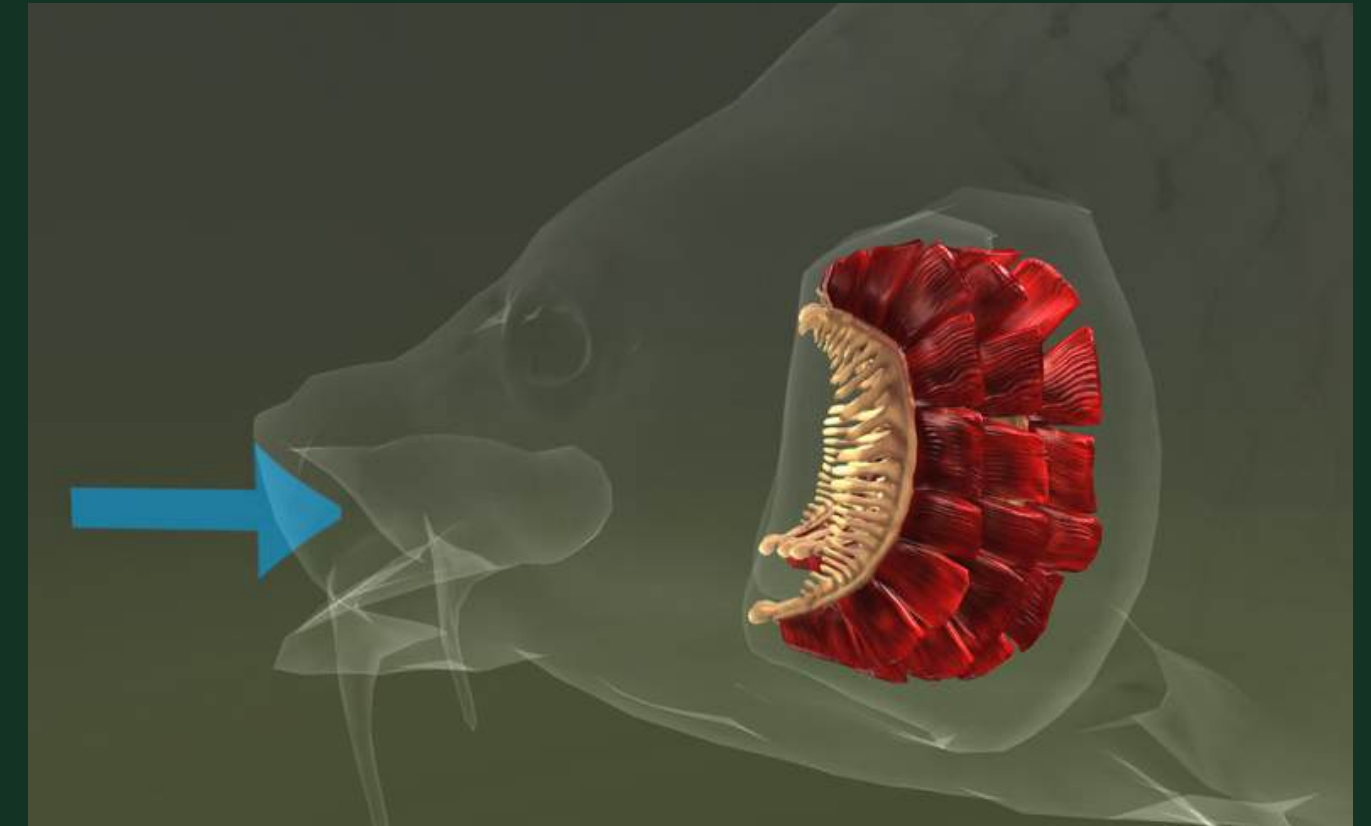
- a) Respiration in plants occurs only during the daytime.**
- b) Oxygen is not required for respiration in plant cells.**
- c) Only leaves carry out respiration in plants.**
- d) Both photosynthesis and respiration occur in leaves during the day.**

HOW DO FISHES BREATHE UNDERWATER?

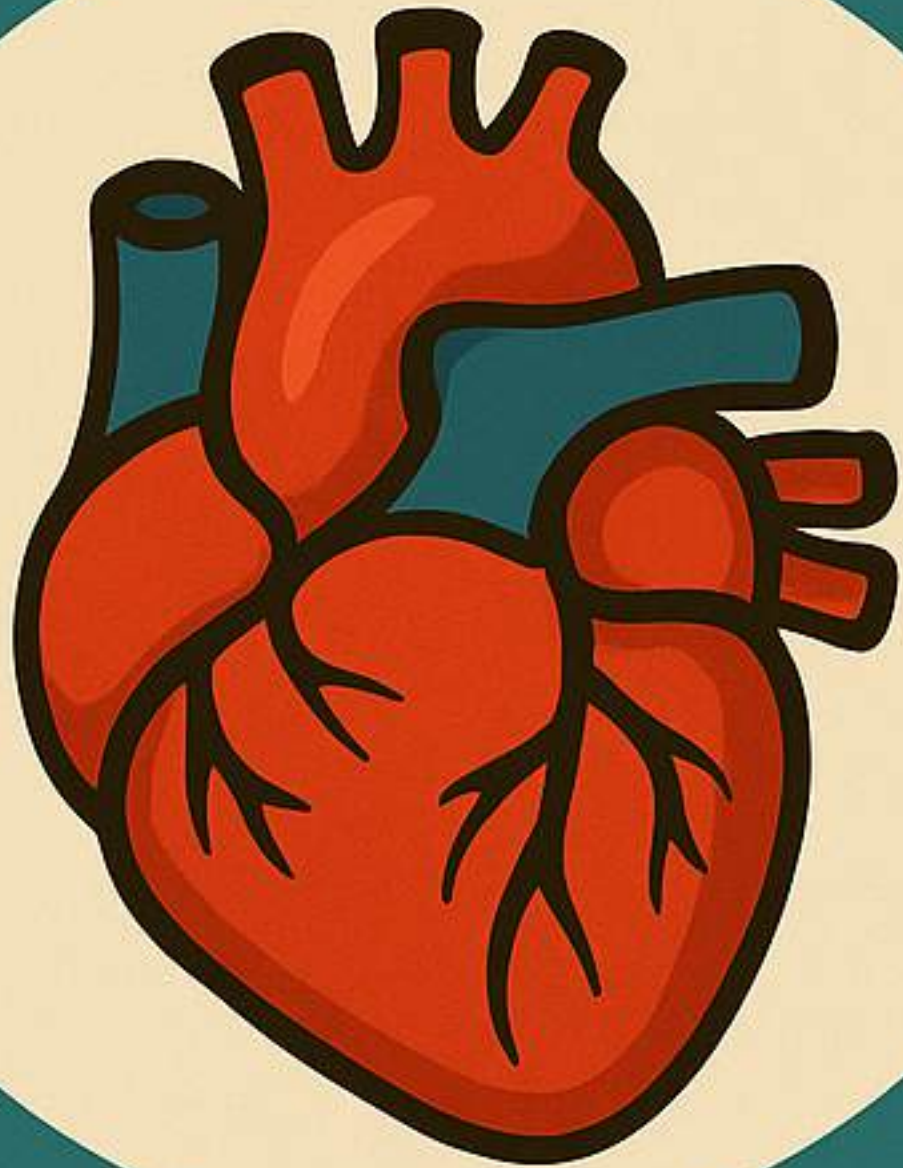
Fishes use gills to absorb dissolved oxygen from water.

Why do aquatic organisms have faster breathing rates than terrestrial organisms?

→ *Because water has less dissolved oxygen than air.*



TRANSPORTATION

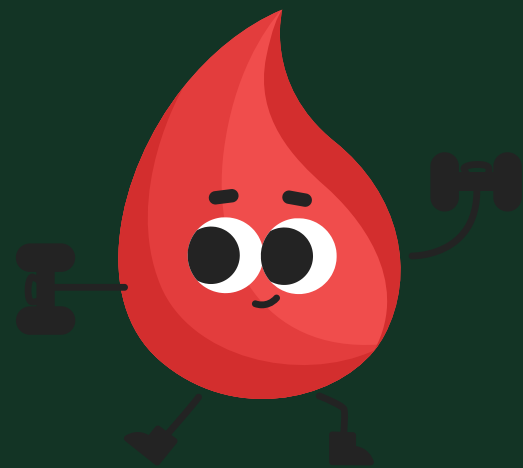


TRANSPORTATION

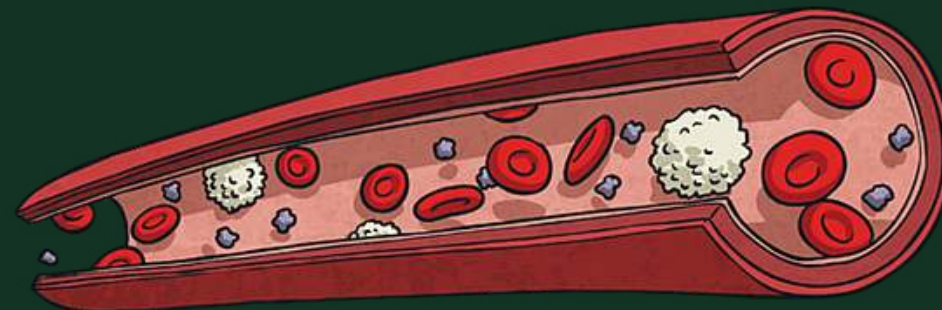
Transportation in living organisms refers to the process of moving food, water, oxygen, and other essential materials to different parts of the body, as well as removing waste products.

Circulatory System

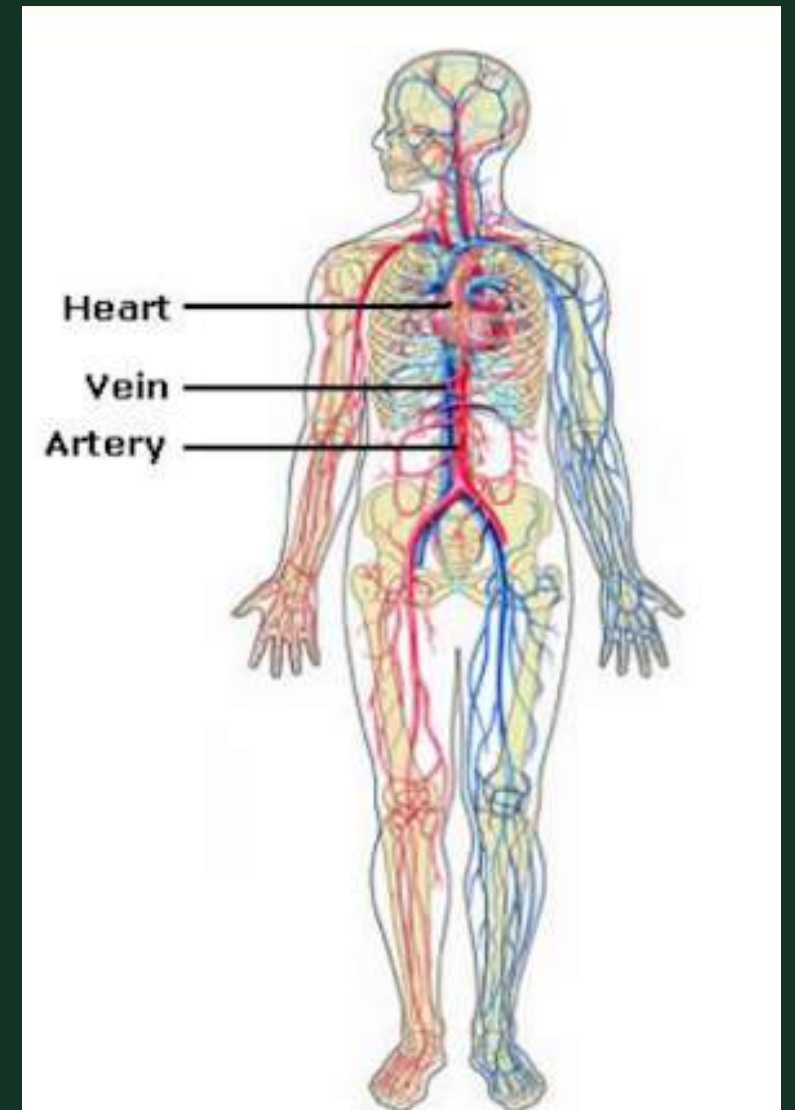
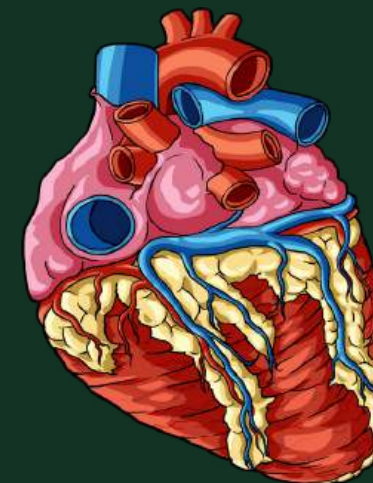
Blood
(Fluid connective tissue)



Blood Vessels
(Tubes)



Heart
(Pumping Organ)



BLOOD



- Blood is a red coloured fluid **connective tissue**, which circulates in our body.
- Its red colour is due to the presence of a respiratory pigment called haemoglobin found in red blood cells (RBCs).
- *The deficiency of haemoglobin in our body is referred as anaemia.*

FUNCTIONS OF BLOOD

- **Transport:** Carries oxygen, nutrients, hormones, and waste.
- **Oxygenation:** RBCs deliver oxygen to body cells.
- **Defense:** WBCs fight infections and destroy germs.
- **Temperature Regulation:** Distributes heat to maintain body temperature.
- **Clotting:** Platelets help stop bleeding by forming clots.

COMPONENTS OF BLOOD

```
graph TD; Root[COMPONENTS OF BLOOD] --> BC[Blood cells]; Root --> P[Plasma]; Root --> PL[Platelets]; BC --> RBC[Red Blood Cells]; BC --> WBC[White Blood Cells];
```

Blood cells

Plasma

Platelets

Fluid medium carrying nutrients, waste products, and other substances. (pale yellowish in color)

Help in blood clotting to prevent leaks and maintain pressure

Red Blood Cells

Contain haemoglobin to transport oxygen and carbon dioxide. (red in color)

White Blood Cells

Provides Immunity, produce antibodies against disease causing microbes

BLOOD VESSELS

Feature	Arteries	Veins	Capillaries
Direction of Blood	Carry blood away from the heart	Carry blood towards the heart	Connect arteries and veins
Oxygen Content	Oxygenated (except pulmonary arteries)	Deoxygenated (except pulmonary veins)	Both (exchange oxygen and carbon dioxide)
Pressure	High	Low	Low
Wall Thickness	Thick and elastic	Thin	Very thin (one cell thick)
Valves	Absent	Present	Absent
Function	Distribute blood pumped by the heart	Return blood to the heart	Exchange oxygen, nutrients, and waste with cells
Carries	Oxygen and nutrients	Carbon dioxide and waste	Oxygen and nutrients to cells, carbon dioxide and waste from cells

“KAAM KI BAAT”

- **Largest Artery:** **Aorta**
- **Only artery that carries deoxygenated blood:** **Pulmonary Artery**
- **Largest Vein:** **Vena Cava**
- **Only vein in that carries oxygenated blood:** **Pulmonary Vein**

Valves in veins help in preventing the backflow of blood.

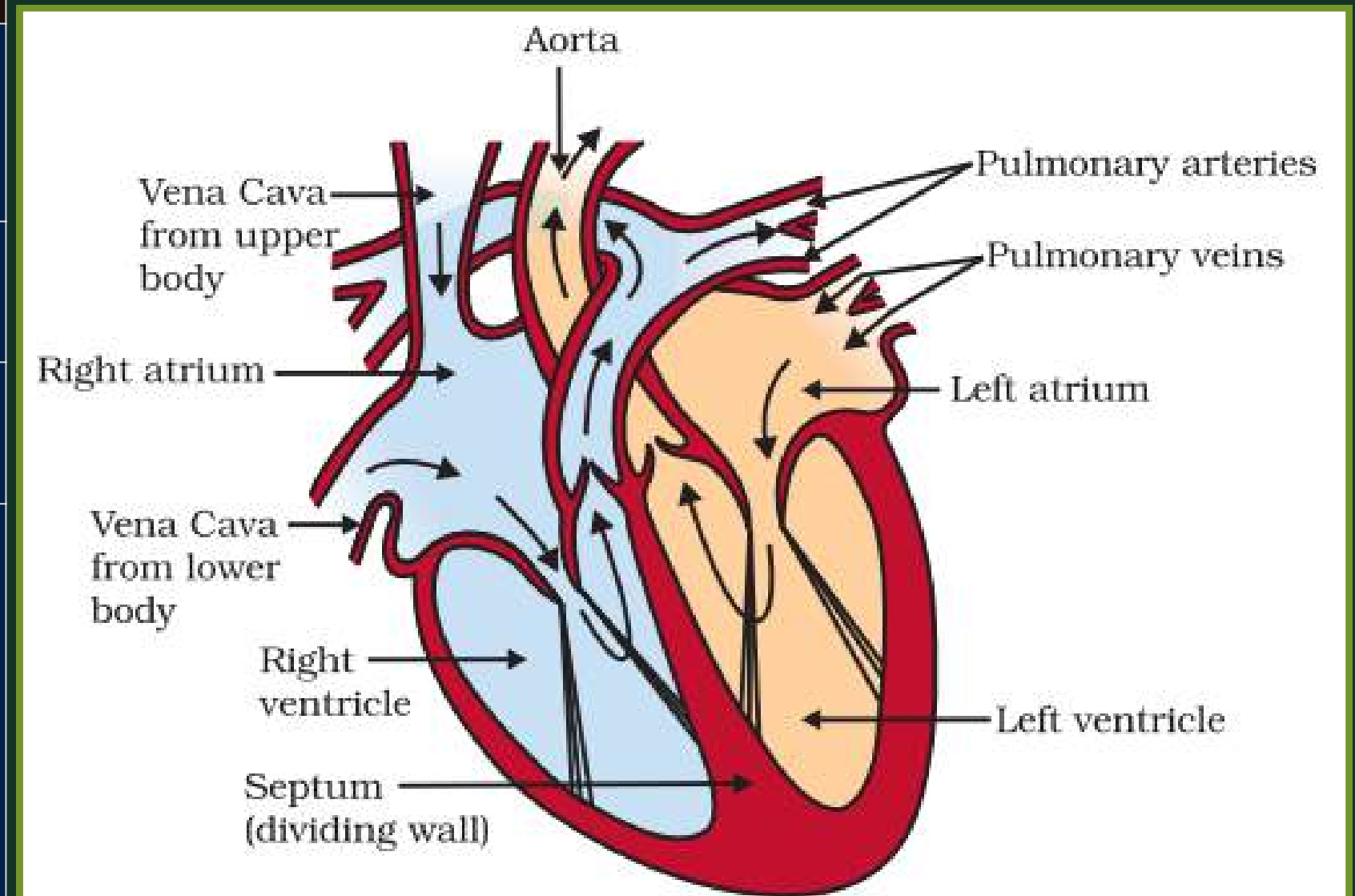
They ensure that blood flows only in one direction – towards the heart, even against gravity, especially from the legs.

This helps maintain proper circulation and avoids blood pooling in one place.

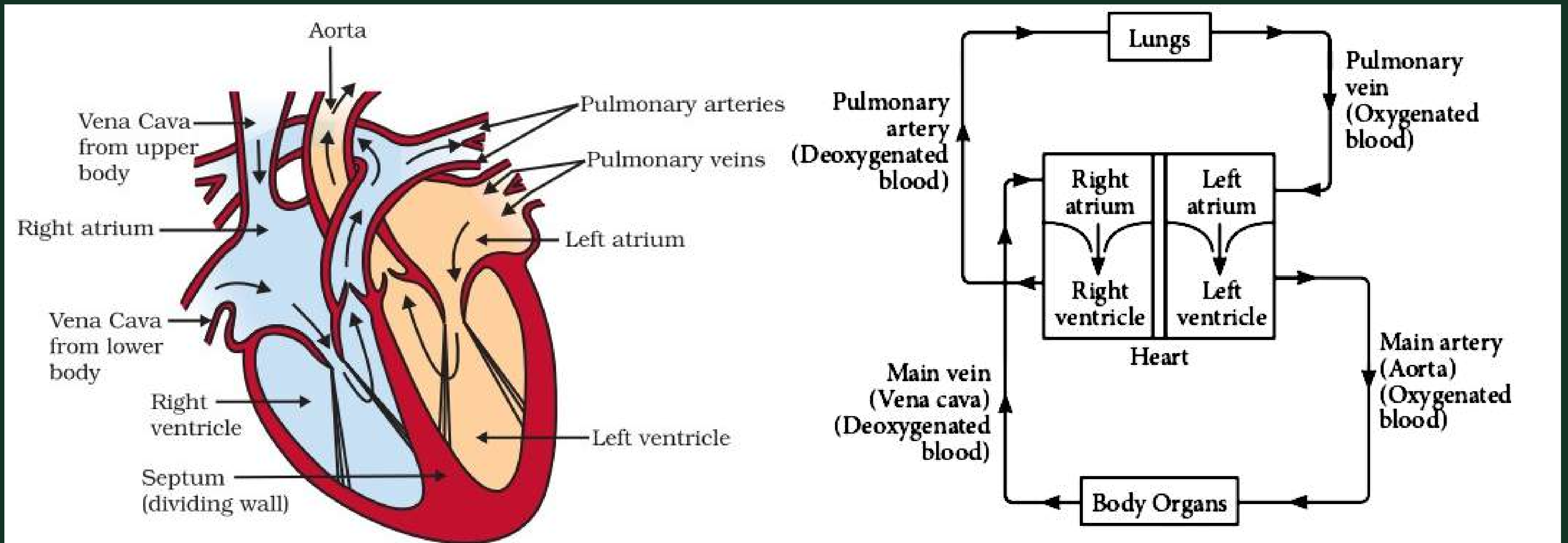
HUMAN HEART

*The heart is a muscular organ that is situated in the front of the chest. It pumps blood all through the body in a process called circulation. It has four chambers separated by **septum** which prevent mixing of pure and impure blood.*

Atrium	Ventricle
Upper chambers of the heart	Lower chambers of the heart
Walls are thin	Walls are thick and muscular
Receive blood from veins	Pump blood into arteries
Right atrium receives deoxygenated blood from body; Left atrium receives oxygenated blood from lungs	Right ventricle pumps deoxygenated blood to lungs; Left ventricle pumps oxygenated blood to the body



HUMAN HEART



Chambers in heart – To prevent mixing of oxygenated and deoxygenated blood. More oxygen will be supplied to body organs in better way.

In humans, blood passes through the heart twice during one complete circulation of the body. This is called double circulation.

Two Types of Circulation:

Pulmonary Circulation: Blood flows from the heart to the lungs and back.

Purpose: To oxygenate the blood (remove CO_2 and take in O_2).

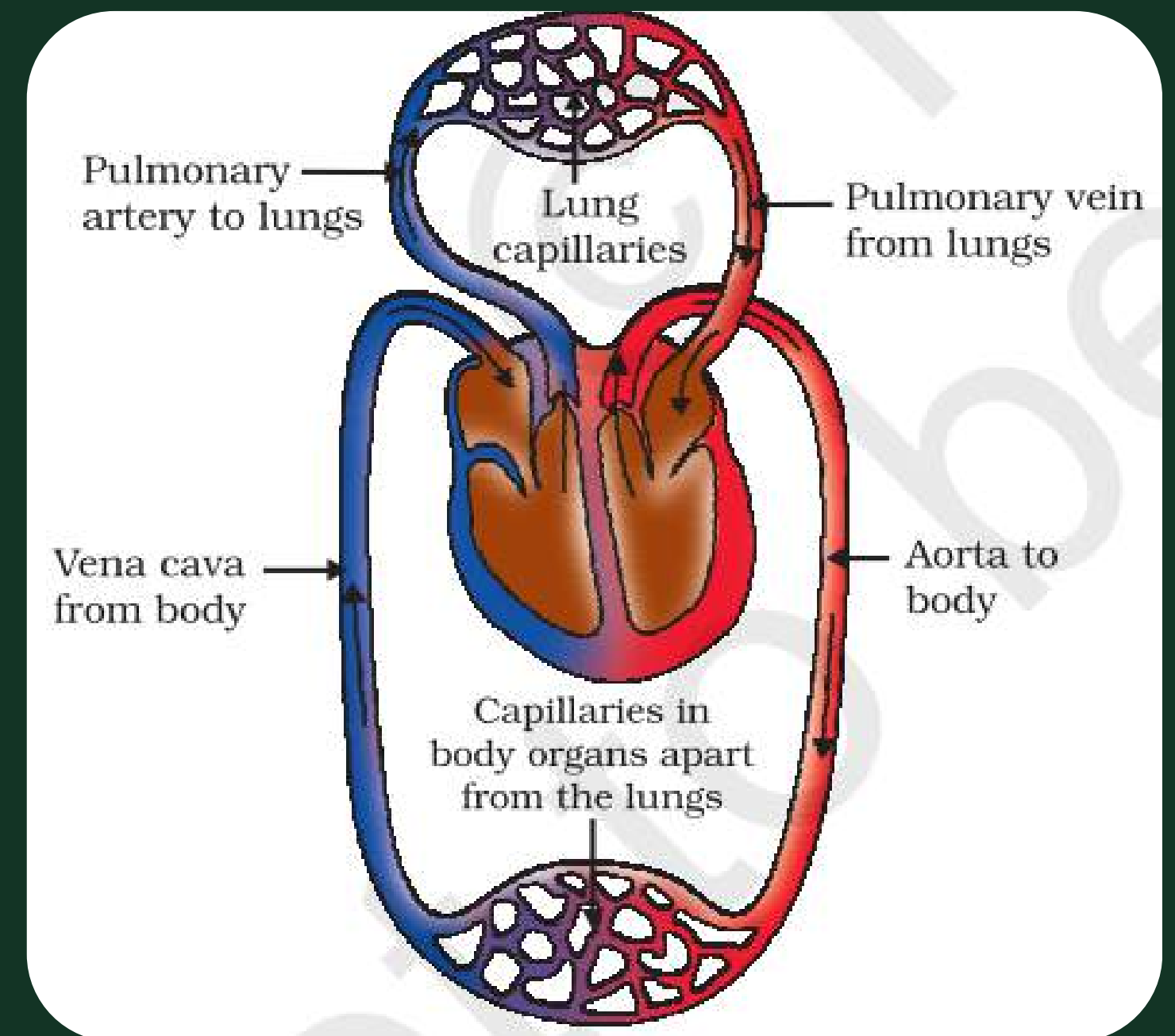
Systemic Circulation: Blood flows from the heart to the rest of the body and returns.

Purpose: To supply oxygen and nutrients to body tissues and collect CO_2 .

Why is it Important?

Keeps oxygen-rich blood separate from oxygen-poor blood. Ensures efficient supply of oxygen to the body.

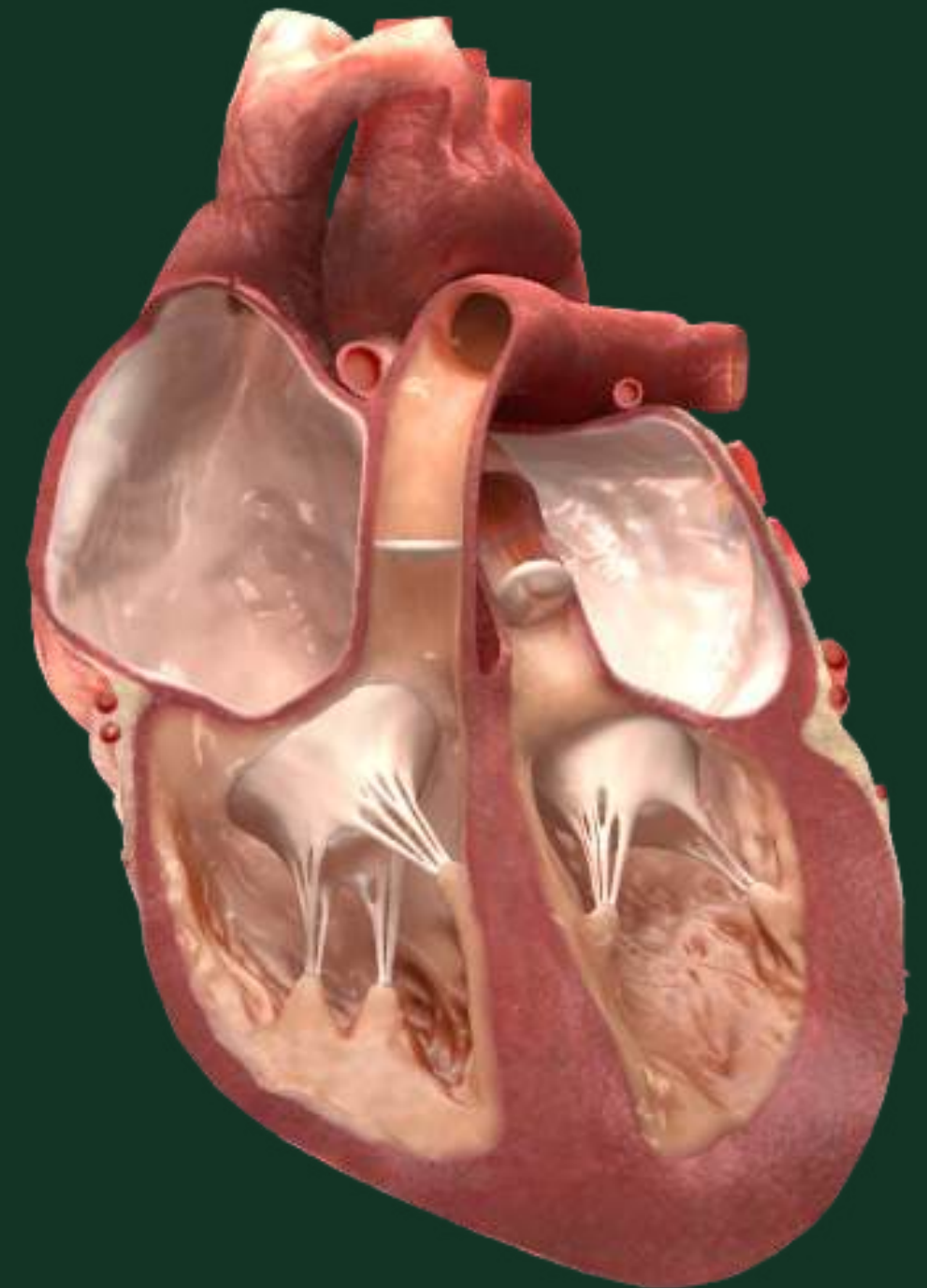
Maintains high pressure in systemic circulation for proper distribution.



**Double
Circulation**

Q. Why is it necessary to have four chambers in the heart of humans and other mammals?

- **Prevents Mixing of Blood:** Oxygenated and deoxygenated blood are kept separate, which is important for efficient circulation.
- **Ensures Efficient Oxygen Supply:** Body gets pure oxygen-rich blood, helping in high energy production.
- **Essential for Warm-Blooded Animals:** Humans need more energy to maintain constant body temperature, so a 4-chambered heart is necessary.



2018, 2020, 2022, 2024 (Practice Paper)

Q. Which component of blood helps in clotting?

- a) Red Blood Cells (RBC)**
- b) White Blood Cells (WBC)**
- c) Platelets**
- d) Plasma**

2020, 2022, 2023, 2025 (Sample & Practice Papers)

Q.: Which of the following statement(s) is (are) true about human heart?

(A) Right atrium receives oxygenated blood from lungs through pulmonary artery.

(B) Left atrium transfers oxygenated blood to left ventricle which sends it to various parts of the body.

(C) Right atrium receives deoxygenated blood through vena cava from upper and lower body.

(D) Left atrium transfers oxygenated blood to aorta which sends it to different parts of the body.

(a) (A)

(b) (A) and (D)

(c) (B) and (C)

(d) (B) and (D)

2021, 2023, 2025 (Sample Paper)

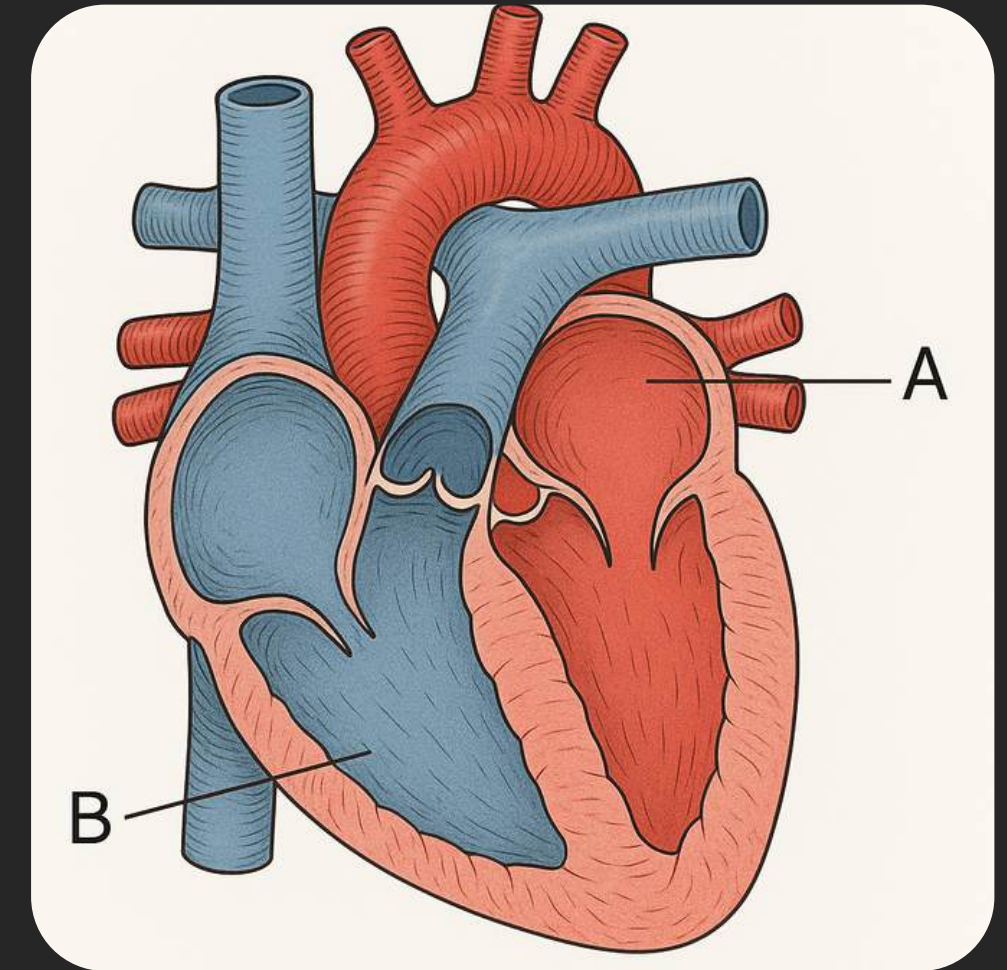
Q. Assertion and Reasons

Assertion (A): The walls of atria are thicker than those of the ventricles.

Reason (R): Ventricles have to pump blood into various organs at high pressure.

2019, 2021, 2022, 2024 (Practice Paper)

Q. Label parts A and B in the diagram of the human heart and write their functions.



2018, 2020, 2022, 2024 (Practice Paper)

Q. Differentiate between pulmonary artery and pulmonary vein based on the type of blood they carry and the direction of blood flow.

<i>Feature</i>	<i>Pulmonary Artery</i>	<i>Pulmonary Vein</i>
<i>Type of Blood</i>	<i>Carries deoxygenated blood</i>	<i>Carries oxygenated blood</i>
<i>Direction of Flow</i>	<i>From heart to lungs</i>	<i>From lungs to heart</i>

2019, 2020, 2022, 2024 (Sample/Practice Papers)

Q. (A) Why are the walls of the ventricles thicker than the walls of the atria?

(B) Why do arteries have thick and elastic walls?

Ans. (A) The walls of the ventricles are thicker than the walls of the atria because ventricles have to pump blood with greater force—right ventricle to the lungs and left ventricle to the entire body—so they need stronger muscular walls.

(B) Arteries have thick and elastic walls because they carry blood from the heart at high pressure. The elasticity helps them withstand and absorb this pressure, preventing rupture and maintaining smooth blood flow

Animal Group	Heart Chambers	Circulation Type	Temperature Control	Blooded Type
 Fishes (Pisces)	2 chambers	Single circulation	No temperature regulation	Cold-blooded
 Amphibians	3 chambers	Incomplete double circulation	No temperature regulation	Cold-blooded
 Reptiles	3 chambers	Incomplete double circulation	No temperature regulation	Cold-blooded
 Birds (Aves)	4 chambers	Complete double circulation	Yes, internal regulation	Warm-blooded
 Mammals	4 chambers	Complete double circulation	Yes, internal regulation	Warm-blooded

Q: Why do more chambers in the heart matter in evolution?

More heart chambers help in better separation of oxygenated and deoxygenated blood. This increases oxygen supply to cells, improves respiration, produces more ATP (energy), and allows warm-blooded animals to maintain body temperature. It makes circulation more efficient, helping organisms survive better.

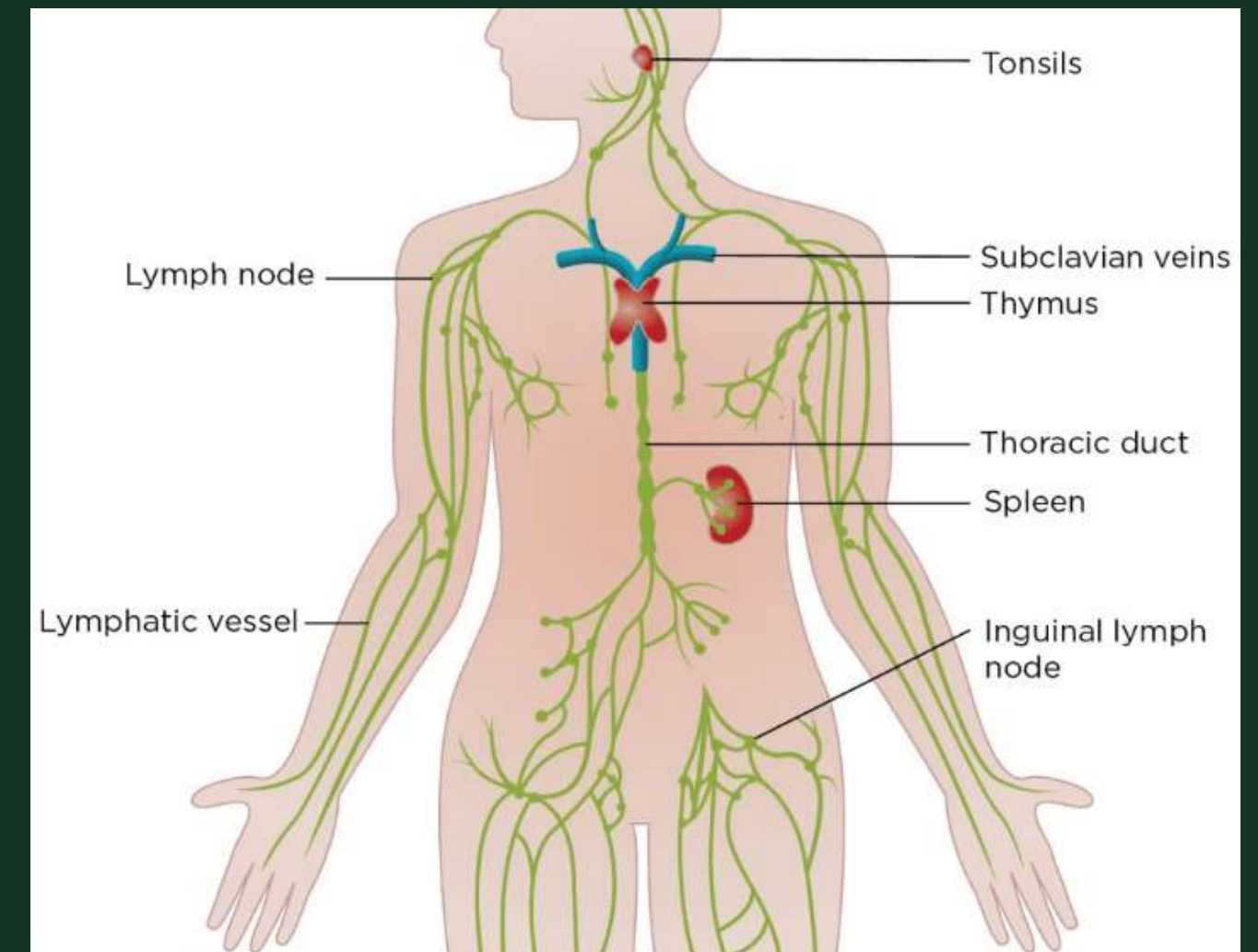
LYMPHATIC SYSTEM

Lymph is also called tissue fluid. Formed when plasma, proteins, and some blood cells escape through the pores of capillary walls into intercellular spaces. It is similar to blood plasma but is: Colourless and contains less protein

- Lymph flows into lymphatic capillaries, which join to form larger lymph vessels.
- These vessels finally open into larger veins, reconnecting to the bloodstream.

Functions of Lymph:

- Transports digested and absorbed fats from the intestine to blood
- Drains excess fluid from tissues (extracellular spaces) back into the blood
- Lymph contains lymphocytes that help defend the body against infections.



2020, 2022, 2024

Q. Which of the following is not a function of the lymphatic system?

- a) Transport of digested fats from the intestine**
- b) Return of excess tissue fluid to the bloodstream**
- c) Oxygen transport to body tissues**
- d) Defense against infections by carrying lymphocytes**

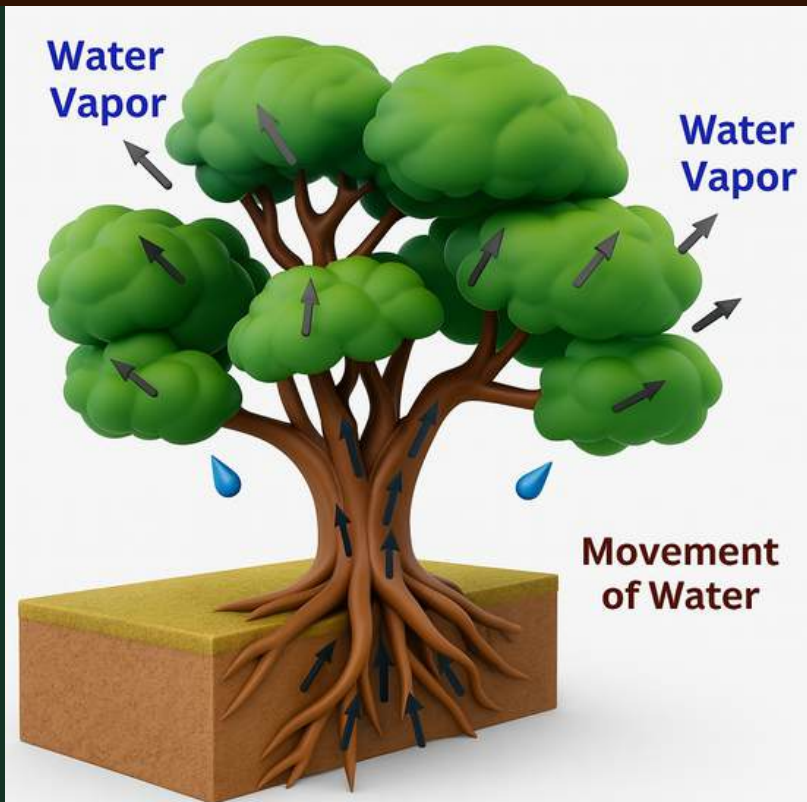
TRANSPORTATION IN PLANTS:

Xylem	Phloem
1. Conducts water and minerals from roots to leaves.	1. Conducts food from leaves to all parts of the plant.
2. Composed of mainly dead elements.	2. Composed of mainly living elements.
3. Transport is unidirectional (only upwards).	3. Transport is bidirectional (both upwards and downwards).
4. Four types of cells: Tracheids, Vessels, Xylem Parenchyma, and Xylem Fibres.	4. Four types of cells: Phloem Fibres, Sieve Tubes, Companion Cells, and Phloem Parenchyma.

TRANSPORTATION IN PLANTS:

Transport of Water and Minerals – by Xylem

Xylem tissues transport water and minerals from roots to leaves. Water enters roots by osmosis and moves upward through xylem vessels. It moves upward through xylem vessels – this upward movement is known as the **ascent of sap**. Movement is due to Root pressure and Transpiration pull (main force)



Transport of Food – by Phloem

Phloem transports prepared food (mainly sucrose) from leaves to other parts of the plant (both upward and downward). This process is called translocation.

- It requires energy (ATP).

Transpiration: The loss of water in the form of water vapour through stomata.

Helps in: Pulling water upwards (transpiration pull), Cooling the plant and Maintaining flow of water and minerals

2019, 2021, 2023, 2025 (Sample Paper)

Q. Which of the following statements correctly explains the transpiration pull in plants?

- a) It is the pressure generated by root hairs to push water upwards.**
- b) It is caused by evaporation of water from the leaf surface, pulling water up through the xylem.**
- c) It is the diffusion of minerals from the soil into the xylem.**
- d) It is the force that pushes glucose from the leaves to the roots.**

2018, 2020, 2022, 2024

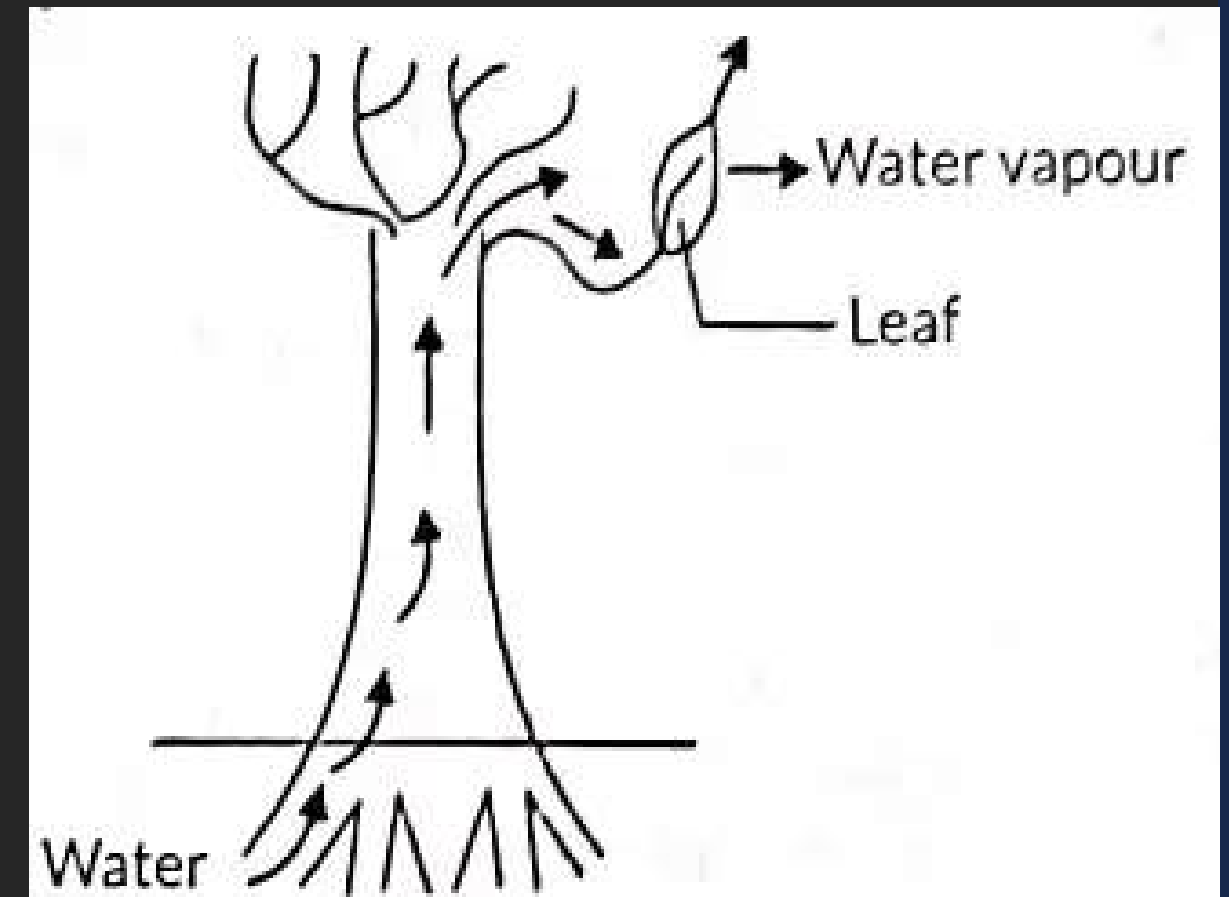
Q. The process in which transport of soluble products of photosynthesis takes place in plants is known as:

- (a) Transpiration**
- (b) Evaporation**
- (c) Conduction**
- (d) Translocation**

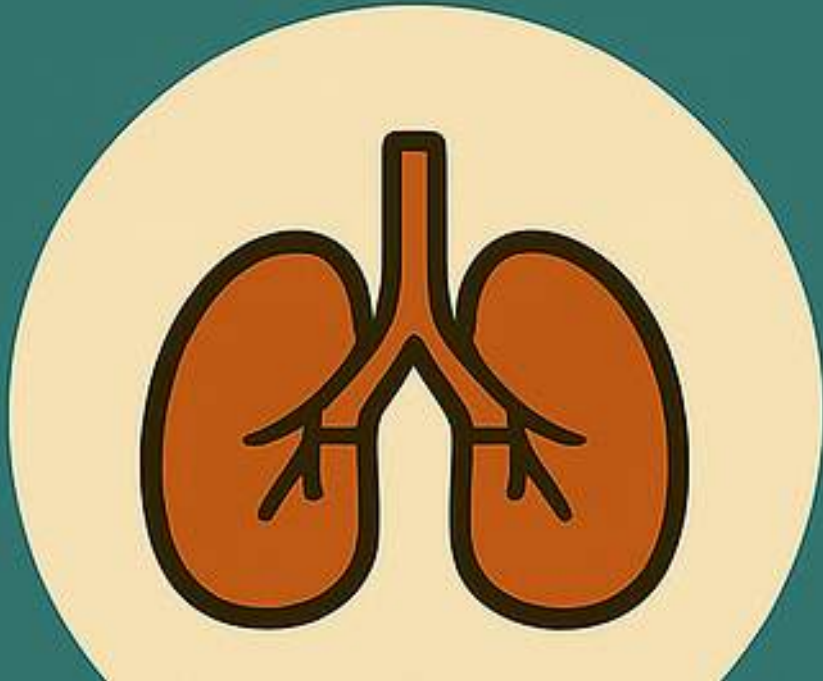
2019, 2021, 2023, 2025

Q. Observe the following diagram and identify the process and its significance from the following options:

- (a) Evaporation : maintains water contents in leaf cells.
- (b) Transpiration : creates a suction force which pulls water inside the plant.
- (c) Excretion : helps in excreting out waste water from the plant.
- (d) Translocation : helps in transporting materials from one cell to another.



EXCRETION

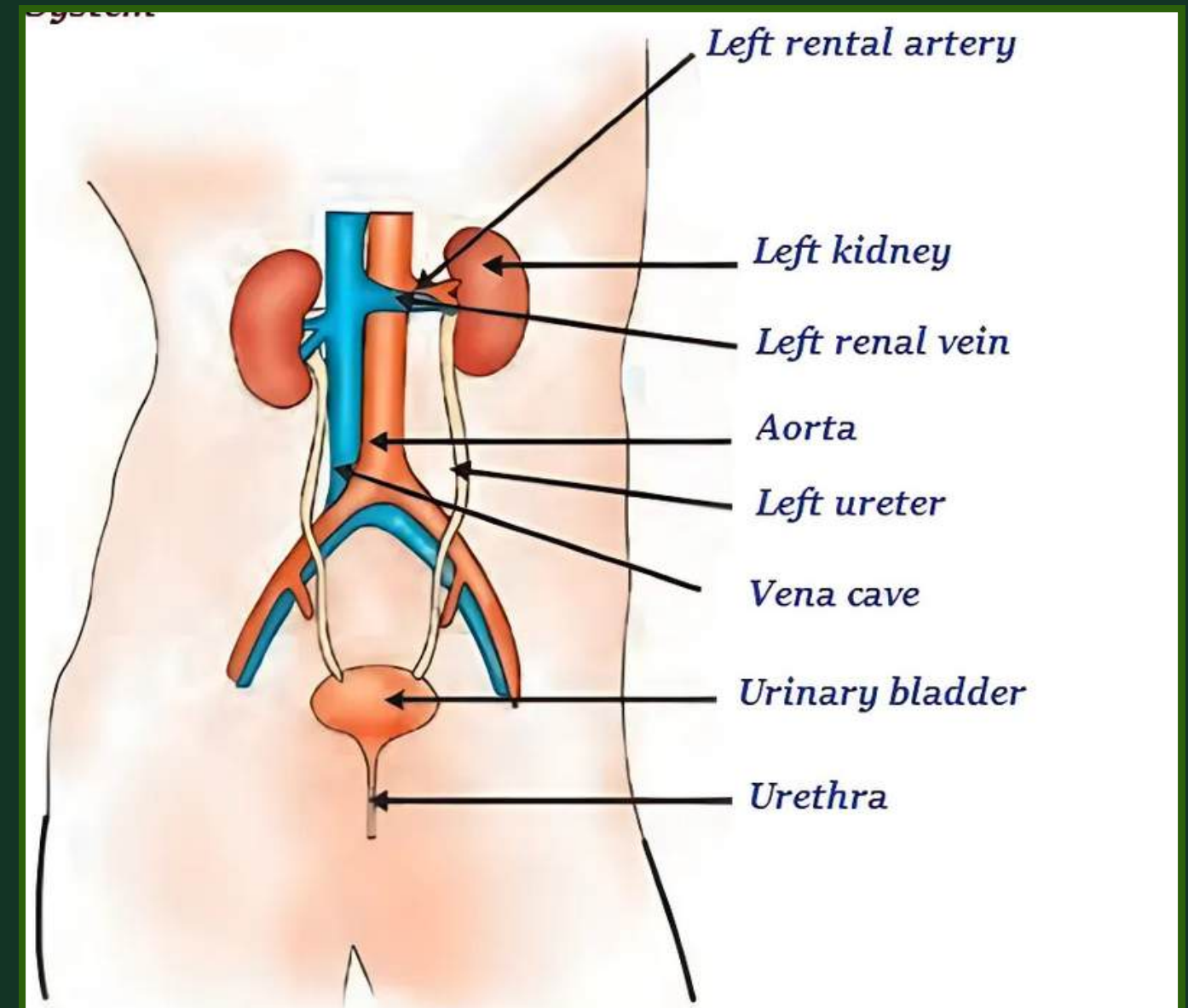


EXCRETION

Excretion is the biological process of removing harmful metabolic wastes (like urea and uric acid) from the body.

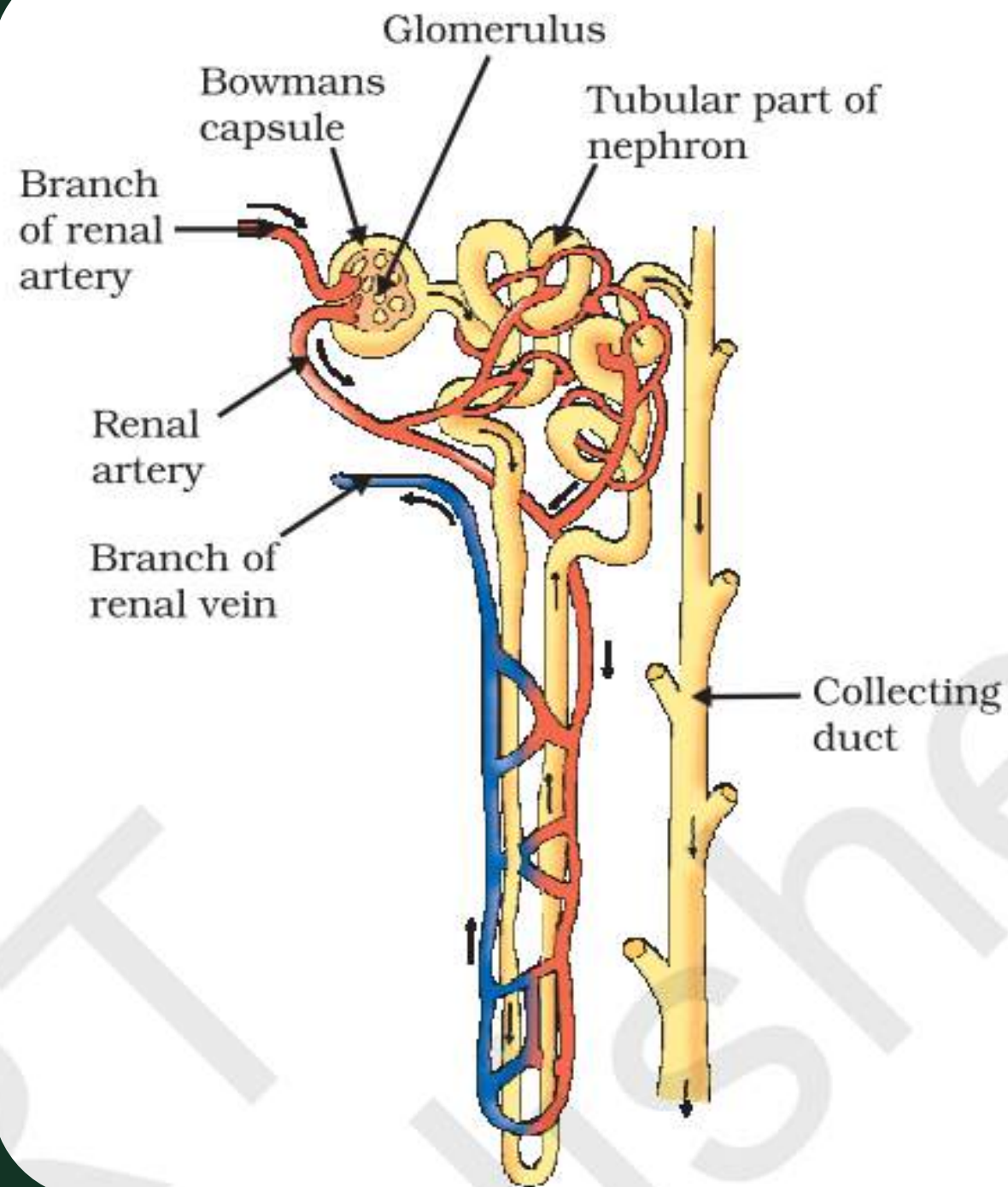
The human excretory system includes:

- **2 Kidneys** – Filter nitrogenous waste from blood to form urine.
- **2 Ureters** – carry urine from kidneys to bladder
- **Urinary Bladder** – stores urine temporarily
- **Urethra** – releases urine out of the body



NEPHRONS

Structural & functional unit of kidney/excretion system. Responsible for filtration of blood. There are millions of nephron in each kidney.



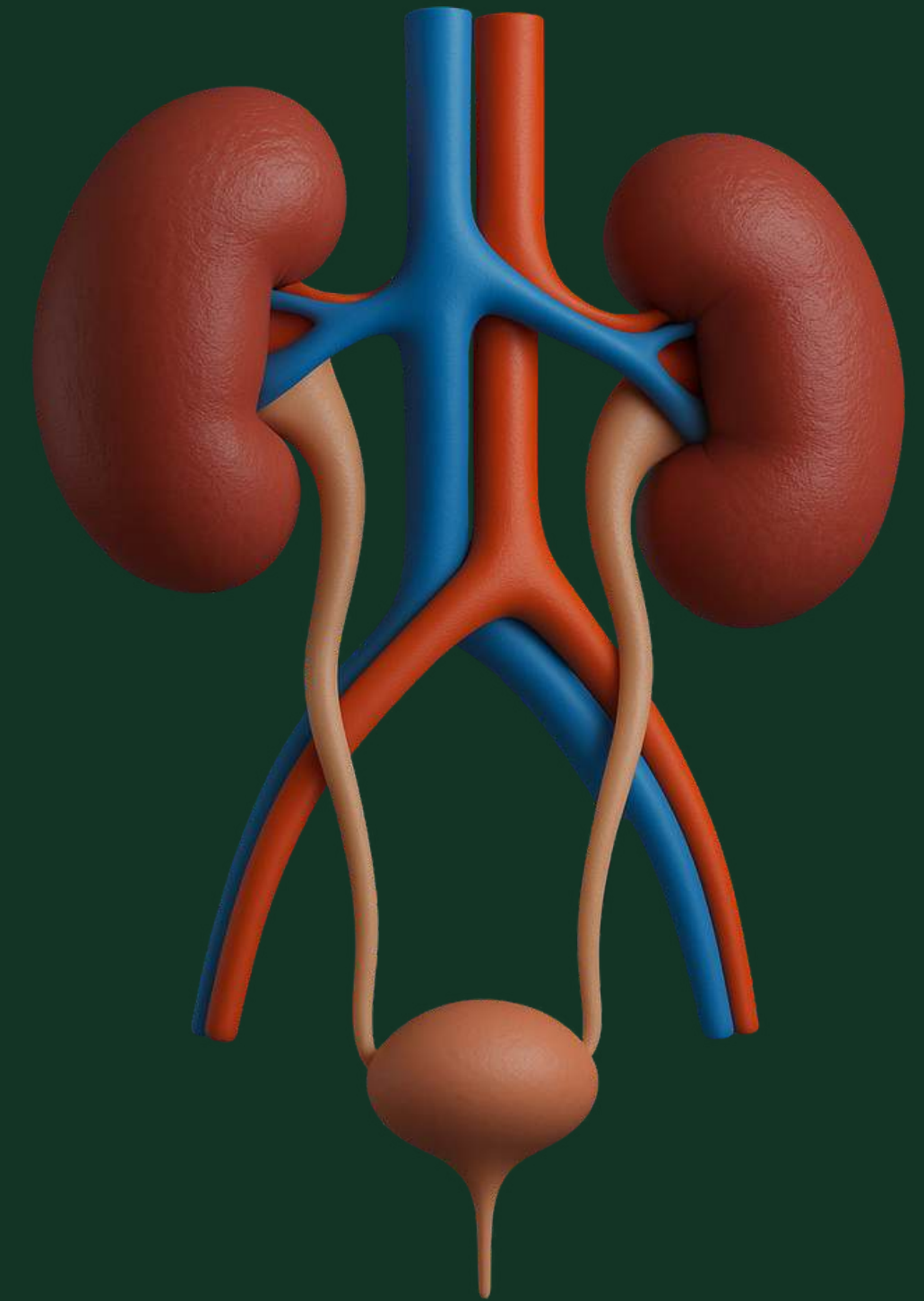
Glomerular Filtration: Blood enters the kidneys, and in the nephrons, nitrogenous wastes, glucose, water, amino acids, and excess salts are filtered out. This initial filtrate enters the Bowman's capsule of each nephron.

Selective Reabsorption: As the filtrate moves through the nephron, useful substances such as glucose, amino acids, salts, and a significant amount of water are reabsorbed back into the blood by the capillaries surrounding the nephron.

Tubular Secretion: Urea, additional water, and excess salts are secreted into the tubule, which then opens into the collecting duct. The urine then flows from the collecting ducts into the ureters.

FORMATION OF URINE

- **Urine Formation:** After filtration and reabsorption, the remaining fluid is urine, containing waste products and extra water.
- **Transport:** Urine travels from the kidneys through the ureters to the urinary bladder.
- **Storage:** Urine is stored in the urinary bladder until it becomes full.
- **Excretion:** When the bladder is full, the nervous system triggers the urge to urinate, and urine is released from the bladder through the urethra and out of the body.



2019, 2021, 2023, 2025 (Sample Paper)

Q. Which part of the nephron is responsible for filtration of blood?

- a) Bowman's capsule**
- b) Loop of Henle**
- c) Collecting duct**
- d) Tubule**

2018, 2020, 2022, 2024 (Practice Paper)

Q.(i) How is urine produced in the human body? Explain the basic process of urine formation.

(ii) State the function of each of the following in the human excretory system:

(a) Kidney

(b) Ureter

(c) Urinary bladder

Ans. (i) Urine is formed in the nephrons of the kidneys through three main steps:

- Filtration: Blood is filtered in the Bowman's capsule. Waste products, water, salts, and glucose pass into the nephron tubule.*
- Reabsorption: Useful substances like glucose, amino acids, and some water are reabsorbed into the blood from the tubule.*
- Secretion: Additional wastes like ions and drugs are secreted into the tubule from the blood.*

The remaining fluid is urine, which contains wastes and excess water.

(ii) (a) Kidney: Filters waste products from blood and forms urine.

(b) Ureter: Carries urine from kidneys to urinary bladder.

(c) Urinary bladder: Temporarily stores urine until it is passed out of the body.

2020, 2022, 2025 (Sample Paper)

Q. Write one specific function of each of the following organs in relation to excretion in human beings:

(i) Renal Artery

(ii) Urethra

(ii) Glomerulus

(iv) Tubular part of nephron

Ans. (i) Renal Artery: Brings oxygenated and impure blood (with nitrogenous wastes) to the kidneys for filtration.

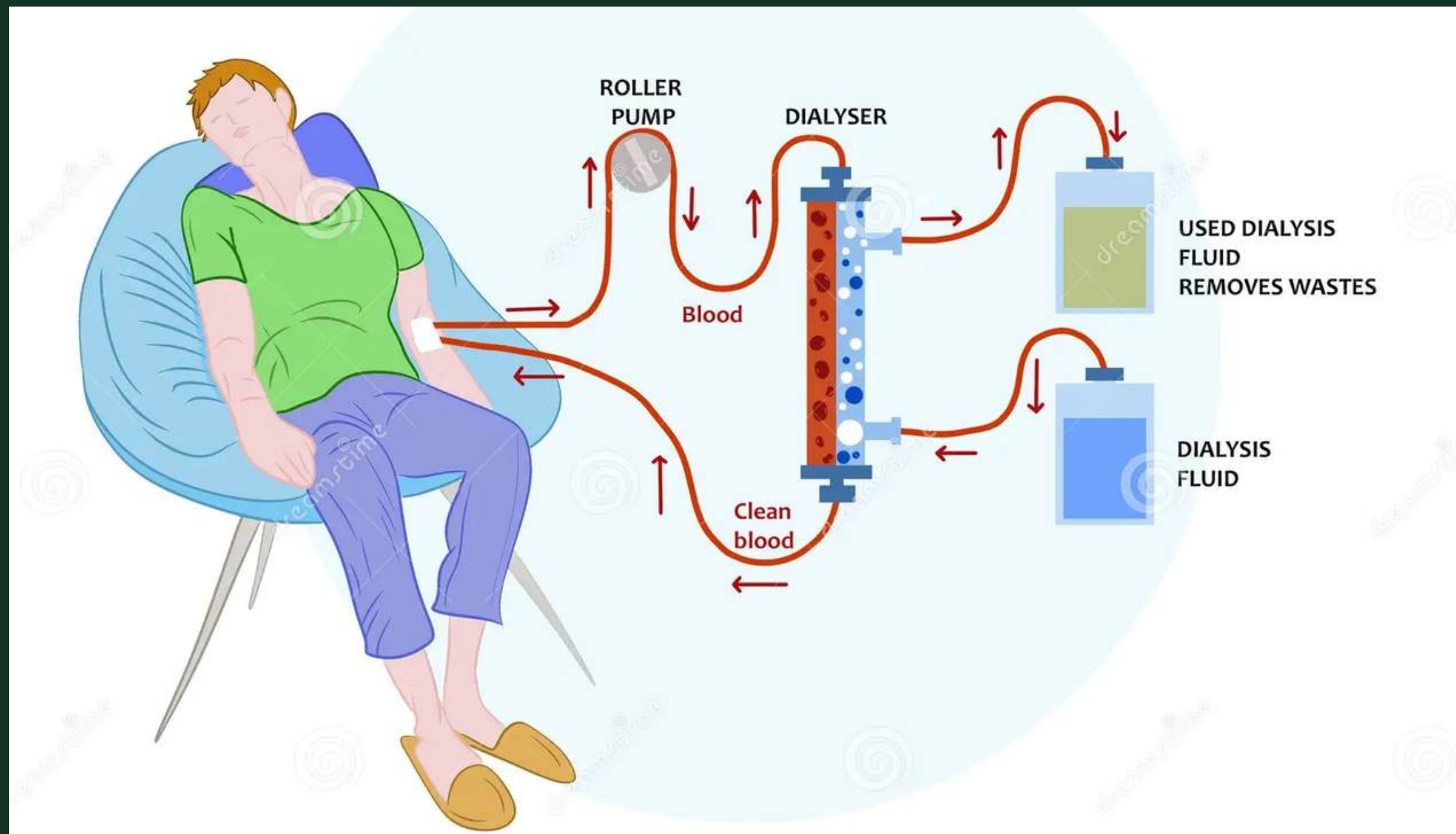
(ii) Urethra: Helps in the removal of urine from the body (urine is passed out through the urethra).

(iii) Glomerulus: Performs ultrafiltration of blood – filters out waste, water, and small molecules into the nephron.

(iv) Tubular Part of Nephron: Reabsorbs useful substances like glucose, amino acids, and water; secretes additional wastes into the filtrate.

HEMODIALYSIS

Hemodialysis is a medical treatment used to filter and cleans the blood of people with kidney failure.



2019, 2021, 2023, 2025

Q. What will happen if kidneys stop working? How is this condition treated in hospitals?

Answer: If kidneys stop working, waste products like urea and excess water start accumulating in the blood, which can be fatal. This condition is called kidney failure or renal failure.

Treatment in hospitals:

- Dialysis – An artificial process to remove waste, excess salts, and water from the blood using a machine (artificial kidney).*
- Kidney Transplant – In severe cases, the damaged kidney is replaced by a healthy kidney from a donor.*

EXCRETION IN PLANTS

Plants use different strategies for excretion of different products :

- Oxygen and carbon dioxide is diffused through stomata.
- Excess water is removed by transpiration.
- Plants can even loose some of their old parts like old leaves and bark of tree.
- Other waste products like raisins and gums especially in old xylem cells which can also be lost by plant
- Plants also secrete some waste substances into the soil around them.



2020, 2022, 2024 (Practice Paper)

Q. Name one gaseous and one non-gaseous waste product excreted by plants. How are these wastes removed?

Ans. Gaseous waste product: Carbon dioxide (CO₂)

Non-gaseous waste product: Resins / Latex / Gum / Excess water / Organic acids



Life is a process – just like cells, your small efforts daily keep you growing. Don't stop.